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Navigating the Amazon: The Incidence of Digital Service Taxes*

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Abstract

Firms in the digital economy often pay little tax in the countries where their customers are based. In response, market countries have introduced digital service taxes on the revenue of these firms to indirectly tax their profits. We study the incidence of these taxes using data on Amazon, the largest online retailer. We find that in most countries, Amazon increased its fees by roughly the amount of the digital service tax. Firms using Amazon as a platform have largely passed these increased fees on to consumers. Large digital firms thus bear only a small part of the tax burden, but the tax may nevertheless succeed in making them less competitive relative to brick-and-mortar retailers.

JEL Classification: H22, D40, L50

Keywords: Tax Incidence, Digital Service Taxes, Two-sided Markets, Platforms

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1 Introduction

Since 2019, several countries have introduced digital service taxes (DSTs). These taxes were initially motivated by concerns that digital firms pay too little profit tax in the countries where they serve customers. In 2018, the European Commission estimated that digital businesses faced an effective tax rate of only 9.5%, compared to 23.2% for traditional firms ([European Commission, 2018](#)). The digital sector relies heavily on intangible assets, which facilitate profit shifting to low-tax jurisdictions. In the absence of a globally coordinated solution—such as the OECD’s Pillar One proposal to reallocate taxing rights toward market countries—governments have turned to DSTs. Unlike corporate income taxes, DSTs are levied on revenues generated from user-based services.

More recently, DSTs have also entered debates on international trade policy. While the United States runs a trade deficit in goods with the European Union, it records a surplus in services. As tariffs apply primarily to goods but not to cross-border services, some European policymakers have considered DSTs as a potential response to U.S. tariffs.¹

These policy discussions raise a key question: who ultimately bears the burden of DSTs? If DSTs resemble corporate income taxes, they may primarily fall on capital owners.² If instead they resemble other revenue-based taxes, such as value-added taxes, they may be largely passed on to consumers (see, e.g., [Benzarti and Carloni, 2019](#); [Benzarti et al., 2020](#)). The applicability of VAT findings to DSTs is not straightforward. The digital economy frequently operates through platform-based models, where digital firms offer platforms for other firms to reach consumers. Tax incidence in such two-sided markets may differ substantially from standard incidence results (see, e.g., [Kind et al., 2010](#); [D’Annunzio et al., 2020](#)). Understanding the incidence of DSTs is therefore key to assessing both their distributional consequences and their role as a policy instrument.

So who bears the burden of a DST? We answer this question using data from Ama-

¹Even before the current trade tensions, the U.S.—home to many large digital firms affected by DSTs—strongly opposed these taxes, arguing that they disproportionately target American companies. In response to their introduction, the U.S. threatened tariffs and launched trade investigations. In July 2020, the U.S. Trade Representative concluded that France’s DST unfairly targeted American firms and announced retaliatory tariffs of 25% on a substantial share of French exports. These tariffs were repeatedly delayed until October 2021, when the U.S. and several European countries (Austria, France, Italy, Spain, and the UK) reached an agreement. Under this agreement, the U.S. withdrew the tariffs, while European countries allowed firms to offset DST payments against future liabilities under Pillar One ([Hines, 2023](#)).

²Empirically, only 40-50% of the incidence of corporate taxes is on capital owners ([Fuest et al., 2018](#); [Suárez Serrato and Zidar, 2016](#)).

zon, the largest online retailer and, by revenue, the largest firm in the digital economy. We find that DSTs are largely passed through to consumers. Amazon increases its platform fees following DST introduction, and third-party sellers pass these increases on to consumers, sometimes more than proportionally. Thus, similar to incidence results for tariffs, the economic burden of DSTs falls primarily on domestic consumers rather than on the targeted foreign digital firms.

In the first part of our paper, we ask how Amazon adjusts the fees it charges third-party sellers for using its platform. To do so, we use information on Fulfillment by Amazon (FBA) fees, i.e., the fees Amazon charges for storing, packing, and delivering goods.³ We have fee information on France, Germany, Italy, Spain, and the United Kingdom. We focus on these countries because they are the only markets with separate national Amazon domains (e.g., amazon.fr, amazon.de) at the time of DST introduction, which allows us to clearly identify domestic transactions. Except for Germany, all of these countries introduced a DST. Using a two-way fixed effects model, we find that a one percentage point increase in the DST raises FBA fees by around 0.5%. Using the estimator by [Callaway and Sant'Anna \(2021\)](#), we find substantial cross-country heterogeneity. In the UK, fees increase by more than the DST. In France and Spain, pass-through is close to full. In Italy, fees rise only slightly and the effect is not statistically significant.

In the second part of our paper, we analyze whether these fee increases are passed on to consumers via higher retail prices. To do so, we use data on prices listed on Amazon, which we obtain from [Keepa.com](#), an independent price tracking website. In a DiD setting, we compare identical products sold by third-party sellers using FBA (and thus subject to the DST) to those sold directly by Amazon (not subject to the DST). We find that sellers largely pass on the higher fees to consumers: prices for FBA-sold products increase by about 2% following the introduction of the DST in France, Spain, and the UK, while we find no significant price response in Italy.

The estimated price effects are economically meaningful. For a representative product, net-of-VAT prices increase by EUR 0.35 in France, EUR 0.36 in Spain, and GBP 0.22 in the United Kingdom. Relating these price changes to the total DST liability generated by a transaction implies that consumers bear between 1.5 and 2.5 euros (or pounds) for each euro (or pound) of tax revenue. This over-shifting likely reflects a combination of rounding practices and adjustments in other Amazon fees. In Italy, where Amazon's fees increase only slightly, the implied price effects are close

³It is also possible for third-party sellers to use Amazon as a pure platform and ship goods directly to consumers. However, information on the selling and referral fees due in this case are not publicly available.

to zero, indicating that the platform absorbs most of the tax, possibly to maintain competitiveness in a less developed market.

We further explore heterogeneity in price responses across products and pricing environments. We find that pass-through is substantially stronger for lower-priced goods, consistent with the fact that a given increase in fees represents a larger relative cost shock for these products. We find weaker price responses for products with highly volatile prices, which likely reflects the use of algorithmic pricing that reacts more strongly to competitors' prices than to cost changes. As an additional robustness check, we implement an instrumental variable approach that uses DST-induced changes in platform fees as a source of exogenous cost variation. The IV estimates confirm substantial pass-through to consumer prices and are consistent with more-than-full pass-through.

Our findings have important implications for policymakers debating the introduction or continuation of DSTs. If the tax is largely passed on to local sellers and consumers, it is not well suited as an indirect tax on large digital firms. At the same time, DSTs closely resemble tariffs on digital services: they raise the cost of cross-border intermediation and are primarily borne by domestic market participants. As with conventional tariffs, their use therefore involves a trade-off between strategic policy objectives and domestic welfare costs. While DSTs may reduce the competitive advantage of digital platforms relative to brick-and-mortar retailers, they do so by increasing prices for consumers and businesses that rely on these platforms.

We contribute to the empirical literature on the incidence of indirect taxes, including value-added taxes, tariffs, and platform-specific transaction taxes. A large body of work on VAT finds substantial but incomplete pass-through to consumer prices, with considerable heterogeneity across settings and firms. Evidence ranges from sector-specific studies, such as Finnish hairdressers (Kosonen, 2015) and French car sales and housing repair services (Carbonnier, 2007), to international panel analyses (Benedek et al., 2020). Firm-level responses vary widely: studying restaurants in Finland and Sweden, Harju et al. (2018) show that some firms pass on tax changes fully, while others do not adjust prices at all. Studies of temporary VAT changes further document asymmetric responses to tax increases and cuts (Benzarti et al., 2020; Fuest et al., 2025).

Closely related, a growing literature studies the incidence of tariffs. A robust finding is that tariffs are largely passed through into import prices, implying that the burden falls primarily on domestic importers (Amiti et al., 2019, 2020; Ma et al., 2021; Feng et al., 2023). However, pass-through to final consumer prices is typically lower, as firms

adjust their margins and domestic distribution costs account for a large share of retail prices (Cavallo et al., 2021, 2025). At the same time, incidence varies across sectors and supply chains, with some studies documenting more-than-full pass-through once downstream markups are taken into account (Flaaen et al., 2020, 2025).

Recent work applies these questions to the digital economy and platform markets. Bibler et al. (2021) show that stricter enforcement of hotel taxes increased prices on Airbnb, while Lassmann et al. (2025) find that corporate tax burdens are passed on to advertising prices on Facebook. Using data from ride-hailing markets, Leccese (2024) and Tarduno (2025) document substantial pass-through of platform taxes to riders, with some evidence of more-than-full pass-through. In the context of cross-border e-commerce, Fang and Ma (2024) analyze the EU VAT reform of 2021 and show that the tax is largely borne by consumers and reduces trade volumes.

Our paper brings these strands together by studying the incidence of DSTs. We provide the first direct evidence on the incidence of DSTs. Studying DSTs is particularly relevant, as these taxes are designed to target large digital platforms but at the same time operate similarly to tariffs on cross-border digital services. This raises the question of whether they effectively tax multinational firms or are instead borne by domestic market participants.

The rest of the paper is structured as follows. Section 2 describes the DSTs in the countries we study and Amazon's fee structure. Section 3 overviews the theoretical literature on tax incidence in two-sided markets before we describe the empirical strategy in Section 4. Section 5 presents the data and Section 6 describes the results. Section 7 discusses incidence, and Section 8 concludes.

2 Institutional Setting

2.1 Digital Service Taxes

DSTs are revenue-based taxes levied on specific digital services provided in a given market country. Unlike the corporate income tax, which applies to profits and requires taxable presence, DSTs are imposed on gross revenues generated from activities such as online intermediation, digital advertising, or user data exploitation. They are ad valorem taxes, typically set at rates of 2–3 percent, and apply only to large multinational firms exceeding both a global revenue threshold (commonly EUR 750 million) and a domestic revenue threshold. DSTs were introduced in response to concerns that highly digitalized firms, whose business models rely heavily on intangible assets and

remote service provision, can generate substantial revenues in market countries while reporting limited taxable profits there. By taxing revenues rather than profits, DSTs aim to secure tax payments in jurisdictions where users are located.

Among OECD countries, France was the first to introduce a DST in July 2019. Italy, Spain, and the United Kingdom followed shortly thereafter.⁴ Table 1 summarizes the main characteristics of the European DSTs we study. All four countries apply a global revenue threshold of EUR 750 million (GBP 500 million in the UK) and levy tax rates of 2–3 percent. Domestic revenue thresholds differ across countries.

The date when the DST law was passed does not always coincide with the date when collection began. In France, the law was passed in July 2019, but collection was initially suspended due to ongoing international negotiations and only began in December 2020, when it became clear that a broader agreement on digital taxation—either through the implementation of Pillar 1 of the OECD’s Base Erosion and Profit Shifting (BEPS) project or an EU-wide DST—was unlikely in the near term. In Spain and Italy, collection started relatively soon after the legislation was passed. In the United Kingdom, by contrast, the DST formally became law with the Finance Act 2020 in July 2020 but applied to revenues accruing from April 1, 2020. The April start date had been announced well in advance and confirmed in the March 2020 Budget, so affected firms were informed about the tax and its design before liabilities began to accrue.⁵ In our empirical analysis, we use the date when collection began, as this is when the tax became economically relevant for affected firms.

Broad-based DSTs are designed to tax revenues linked to user participation in a given market country. The underlying rationale is that digital firms derive location-specific value from users through intermediation, advertising, and data collection.⁶ Accordingly, the tax base includes revenues from specified digital services. In France and Italy, the DST applies to online advertising, the sale of user data, and digital

⁴Among OECD members, Turkey also adopted a broad-based DST. We do not include Turkey in our analysis because comparable Amazon marketplace data are not available. Austria and Portugal introduced DSTs with a narrower scope, targeting primarily online advertising revenues. Several non-OECD countries, including India, Kenya, Kyrgyzstan, Nepal, and Tanzania, have also implemented broad-based DSTs. Our analysis focuses on European countries with broad-based DSTs and available Amazon marketplace data.

⁵Under UK fiscal procedure, the House of Commons adopts “Ways and Means” resolutions at the Budget that authorize the provisional collection of taxes before the Finance Act is enacted. The subsequent statute codifies the measure in permanent legislation. The earlier effective date of the DST was chosen to limit forestalling and revenue shifting by digital platforms. The COVID-19 pandemic delayed parliamentary passage of the Finance Act but did not alter the pre-announced start date.

⁶Cui (2019); Cui and Hashimzade (2019) argue that DSTs are better placed to tax location-specific rents from the user base than other proposals to tax the digital sector (such as significant digital presence proposals).

Table 1: Digital service tax characteristics in France, Italy, Spain & UK

Country	Date Law Passed	Tax Rate	Global Rev. Threshold	Dom. Rev. Threshold	Date Implemented
France	Jul 24, 2019	3%	€ 750 mil	€ 25 mil	Dec 1, 2020
Italy	Dec 19, 2019	3%	€ 750 mil	€ 5.5 mil	Jan 1, 2020
Spain	Oct 7, 2020	3%	€ 750 mil	€ 3 mil	Jan 16, 2021
UK	Jul 16, 2020	2%	£500 mil	£25 mil	Apr 1, 2020

Notes: This table provides an overview of the DSTs in the countries studied. “Date Law Passed” refers to when the DST legislation was approved. The “Global Rev. Threshold” indicates the minimum worldwide revenues a firm must have to be subject to the DST, while the “Dom. Rev. Threshold” specifies the minimum revenues within the respective country. Both thresholds must be met for the DST to apply. “Date Implemented” is when DST collection began. *Source:* [KPMG \(2024\)](#).

intermediary services such as online marketplaces. Spain’s DST is similar but excludes user data sales. The UK’s DST is somewhat narrower and covers revenues from search engines, social media platforms, and online marketplaces. For Amazon, this implies that fees charged to third-party sellers for using its marketplace are subject to the DST, whereas revenues from Amazon’s direct retail sales are not. This distinction is central to our empirical design, as it allows us to compare marketplace transactions that fall within the tax base to direct sales that do not.

Due to their limited scope, DST revenues are relatively small, ranging from EUR 279 million in Spain to GBP 713 million in the UK in 2022, amounting to roughly 0.1% of total tax revenues ([OECD, 2024](#)). As the digital economy is growing rapidly, there is some scope for increases in revenue. Notably, the revenue comes from a very small number of firms. In the UK, approximately 90% of total DST revenue was generated by just five companies ([National Audit Office, 2022](#)).⁷

For comparison, estimated revenue losses from international profit shifting are substantially larger. France is estimated to have lost around EUR 13 billion in corporate tax revenue due to profit shifting in 2021, the United Kingdom around EUR 19 billion, and Spain approximately EUR 5 billion ([Tørsløv et al., 2023](#)). While such estimates are subject to uncertainty, they exceed annual DST revenues several-fold. Also revenues from tariffs on goods substantially exceed DST revenues: customs duties collected by the European Union amounted to EUR 26.8 billion in 2024. Thus, relative to both estimated profit-shifting losses and tariff revenues, DST revenues are small in aggregate terms. Their economic relevance therefore lies less in their contribution to total revenues

⁷In 2020-21, 18 business groups paid DST in the UK. Interestingly, eleven of the 20 business groups initially expected to make a payment determined that they did not owe any DST ([National Audit Office, 2022](#)).

and more in their distributional and competitive effects within digital markets.

2.2 Amazon's Business Model and Fee Structure

Amazon operates a dual business model. It acts both as a direct retailer, selling products to consumers on its own account, and as a platform that enables third-party sellers to reach consumers.⁸ In 2023, total worldwide sales by Amazon and sellers on its platform reached USD 700 billion. Of this amount, USD 480 billion were generated by third-party marketplace sales—more than double Amazon's first-party sales of USD 220 billion (Kaziukėnas, 2024). Almost 70% of these sales occurred in the United States. Among European countries, Germany (USD 38 billion in 2023) and the United Kingdom (USD 34 billion) are the largest markets. Penetration rates—the proportion of consumers who have made a purchase among potential consumers—are similarly high across all countries we study, ranging from 50% in Spain and 52% in France to 58% in the UK and 61% in Italy (NielsenIQ, 2024).

Third-party sellers pay several types of fees for using Amazon's platform. The primary fee is the *referral fee*, which is charged as a percentage of the total sales price and varies by product category. For many categories, the referral fee is around 15%, with exact charges varying by product category, and, in some cases, price tier. Sellers who use Amazon's *Fulfillment by Amazon (FBA)* service, where Amazon manages storage, packaging, and shipping, are also subject to *FBA fees*. These fees depend on the size and weight of the item, with larger or bulkier products incurring higher charges. In addition, sellers may face *storage fees* if products remain in Amazon's warehouses for extended periods, as well as *optional service fees*, such as advertising expenditures or multi-channel fulfillment fees for orders placed on other platforms but fulfilled by Amazon.⁹

In all countries we study, all marketplace-related fees fall within the DST base. By contrast, revenues from Amazon's direct retail sales are not subject to the DST. This distinction is central to our empirical design. Our analysis focuses on FBA fees, as only on these data are publicly observable over time.

⁸Amazon also operates other important business lines, most notably Amazon Web Services (AWS). Our analysis does not consider these activities. Cloud-computing revenues are generally not within the scope of the DSTs.

⁹Regulatory authorities in both the United States and the EU have scrutinized Amazon's fee structure. The Federal Trade Commission (FTC) has investigated whether aspects of Amazon's pricing and fee policies may disadvantage smaller sellers. In the EU, the Digital Markets Act (DMA) designates Amazon as a "gatekeeper" and imposes additional obligations aimed at ensuring fair competition and transparency in digital markets.

3 Tax Incidence in Two-Sided Markets

Studying tax incidence implies asking who ultimately bears the economic burden of a tax. The core idea is that the legal responsibility to pay a tax (the statutory incidence) may not align with the true economic impact (the economic incidence). This discrepancy arises because taxes alter behavior, influencing equilibrium prices and shifting the burden between buyers and sellers. In basic partial-equilibrium models, the distribution of the tax burden depends on the relative elasticities of supply and demand (see [Fullerton and Metcalf, 2002](#), for a review). When demand is less elastic than supply, consumers bear a larger share of the tax burden because they are less responsive to price changes. Conversely, when supply is less elastic than demand, producers shoulder more of the tax burden because they are less able to adjust their output.

Market structure is an important determinant of pass-through. In perfectly competitive markets with constant marginal costs, ad valorem taxes are typically fully passed on to consumers. Since prices equal marginal costs and firms operate at minimum average cost, firms cannot lower prices to absorb the tax without incurring losses. Under monopoly or oligopoly, by contrast, pass-through depends not only on demand elasticities but also on demand curvature and firms' strategic pricing behavior ([Weyl and Fabinger, 2013](#)). Thus, the extent to which a tax is reflected in consumer prices is an empirical question that depends on market structure and demand elasticities.

In a two-sided market such as Amazon's marketplace, additional effects arise. Amazon acts as an intermediary platform connecting two distinct groups—sellers and consumers. The value of the platform to each group depends on the size and activity of the other: sellers are attracted by a large consumer base, while consumers benefit from a wide variety of sellers. This interdependence creates challenges for analyzing tax incidence because a tax on one side of the market can affect the behavior and outcomes on both sides. Standard models of tax incidence, which assume independent supply and demand curves, can thus provide misleading results.

[Kind et al. \(2010\)](#) model tax incidence in a two-sided platform. They show that an increase in an ad valorem tax on one side of the market affects the relative profitability of the two sides of the platform. Therefore, the firm will want to shift its earnings to the untaxed side. By doing so it reduces the burden of the tax increase. Contrary to what one might expect, this may involve increasing output and reducing prices on both sides of the market. [Kind and Koethenbuerger \(2018\)](#) show that these findings also apply to platforms that do not charge one side of the market. Taking into account that digital companies typically sell services, like advertising, through affiliates in low-tax countries,

Kind et al. (2024) study how a DST affects consumer prices. They find that a DST can intensify competition with non-digital retailers and thus lower consumer prices.¹⁰

These insights are directly relevant for Amazon. The Amazon marketplace is a two-sided platform that generates revenue primarily from fees charged to sellers and, to some extent, from customer memberships like Amazon Prime. The attractiveness of the platform to consumers is positively related to the number of sellers, and vice versa. Kind et al. (2010) suggest that a DST on the fees Amazon charges to sellers may prompt the platform to rely more on income from customers, e.g., by trying to make it more attractive to sign up to Amazon Prime. To counteract the tax impact on sellers, Amazon might reduce its fees to attract more sellers, thereby enhancing the platform's appeal to consumers. This strategy could lead to an increase in both the number of sellers and consumer subscriptions to Amazon Prime.¹¹

To examine whether Amazon shifted the DST burden toward the consumer side of the platform, we analyze the evolution of Amazon Prime membership fees in the countries under study. Around the period we study, Prime fees were adjusted only once, in August 2022, and this change occurred simultaneously across all countries (France, Spain, the UK, Italy, and Germany). The timing does not coincide with the introduction of DSTs in individual countries. We therefore find no evidence that Amazon responded to the DST by adjusting subscription prices. This suggests that any tax-induced adjustment must operate through third-party-seller's profits, Amazon's own margins, or consumer prices for goods sold on the platform.

We turn next to these channels. First, we examine whether Amazon increased marketplace fees in response to the DST. We then analyze whether third-party sellers passed on any fee increases to final consumers.

¹⁰Belleflamme and Toulemonde (2018) analyze the effect of taxes on competing two-sided platforms. The authors use a Hotelling competition model, focusing on how platforms set access prices for two distinct groups of agents who benefit from interacting with each other. They also show that ad-valorem taxes may benefit users that are taxed, but hurts agents on the other side of the market. In a somewhat similar vein, D'Annunzio et al. (2020) study ad-valorem taxes in settings where firms charge multi-part tariffs. Some of these findings can be applied to Amazon's complex fee structure. They also find that ad-valorem taxes can, in some settings, correct underprovision, but only if the government applies differentiated tax rates to different parts of the tariff—a condition not met by the uniform DST. See Sanchez-Cartas and León (2021) for an overview of the theoretical literature on multisided platforms.

¹¹Kind et al. (2008) ask whether taxes can be efficiency enhancing in such a setting. Whether a DST can increase efficiency depends on whether firms over- or underprovide their goods in the absence of the tax. For a monopolistic platform, underprovision can be alleviated by charging a positive ad-valorem tax, due to its counterintuitive effect on quantities described above.

4 Empirical Strategy

4.1 Pass Through to Sellers on Amazon Marketplace

To examine whether Amazon adjusted its FBA fees in response to the introduction of DSTs, we use a staggered DiD design that exploits variation in the timing of DST adoption across countries. Each DST introduction constitutes a treatment event. Countries that have not yet introduced a DST serve as the control group in a given period. Germany, which has not introduced a DST, remains part of the control group throughout.

Our baseline regression equation is

$$\ln(\text{FBA fee})_{jct} = \alpha_1 \text{DST}_{ct} + \alpha_2 X_{ct} + \delta_{jc} + \delta_t + \epsilon_{jct}. \quad (1)$$

The dependent variable, $\ln(\text{FBA fee})_{jct}$, is the natural logarithm of the FBA fee for a size, weight, product- and packaging-category tuple j in country c at time t . The key explanatory variable, DST_{ct} , captures the presence of a DST in country c at time t . We operationalize this variable in two ways: first, as a continuous measure reflecting the DST rate in percent, and second, as a binary indicator that takes the value of one when the DST is in effect and zero otherwise. In some specifications, we include control variables X_{ct} , comprising inflation, the share of digital sales in all retail sales, and a measure for covid policy stringency (see Table B.1 for variable definitions and sources). We add fixed effects δ_{jc} for each fee category j in each country c to control for time-invariant characteristics, and month fixed effects δ_t to account for common temporal shocks.

Defining the treatment timing is challenging here, as there are often substantial delays between when a DST law is passed and when collection starts (see Table 1). For instance, France delayed collection by over a year to allow for international tax negotiations. In the UK, the tax was collected before the law was officially passed; guidance on the expected implementation was published two weeks before collection started. To reflect the actual economic impact, we use the date the tax was first collected as the treatment start date, rather than the legislative date.¹²

The staggered introduction of DSTs across countries raises well-known concerns for traditional two-way fixed effects (TWFE) estimators when treatment timing varies across units. As shown by [Goodman-Bacon \(2021\)](#), TWFE estimates may implic-

¹²In untabulated analyses, we exclude the period between announcement and implementation. The results are similar: coefficient estimates are sometimes somewhat smaller but remain statistically significant.

itly rely on comparisons between already-treated and newly treated units and can assign negative weights to certain group-time cells. To assess the relevance of these issues in our setting, we perform a Bacon decomposition (Table B.2 and Figure B.1 in the Appendix). The decomposition shows that comparisons between treatment-timing groups account for 53% of the identifying variation but contribute a near-zero estimate, whereas treated-versus-never-treated comparisons account for 47% of the identifying variation and yield a substantially larger estimate. This pattern suggests that the TWFE estimate is attenuated by comparisons across treatment-timing groups. We therefore also report modern staggered DiD estimators that are robust to heterogeneous treatment timing (Sun and Abraham, 2021; Borusyak et al., 2024; Callaway and Sant’Anna, 2021). We also present event-study results based on the imputation estimator by Borusyak et al. (2024) and report event studies with additional estimators in the appendix.

The fees for the different size, weight, product- and packaging-category tuples are of different importance for sellers. We therefore apply weights in our regression. The weights are based on the frequency of each fee category in the price dataset, so that more commonly used categories receive higher weight in the estimation.¹³

We cluster standard errors at the cross-sectional identifier (size, weight, product- and packaging-category tuple *times* country level, with around 230 clusters). This matches both the level at which we construct the weights and the level at which the staggered DiD estimators operate. Clustering at the country level is infeasible with only one never-treated country. As a robustness check, we also cluster at country *times* product-type (15 clusters).

4.2 Pass Through to Consumers

We then examine whether third-party sellers on Amazon pass increases in platform fees due to DSTs on to consumers. Our main approach is again a DiD analysis. As a robustness check, we also implement an instrumental variable approach.

To estimate the extent to which DST-induced changes in Amazon’s fees are passed on to consumers, we compare the evolution of prices for identical products sold on Amazon by different sellers. The treated group consists of products sold by third-party sellers using FBA. These sellers pay Amazon fulfillment and referral fees, which are subject to the DST. The control group consists of the same products, but sold directly

¹³Some fee categories do not appear in the price data for certain countries, which leads to slight differences in the number of observations across countries.

by Amazon as the retailer. Since Amazon's direct sales revenue is not subject to the DST, prices in this group should not be directly affected by the tax. Our identifying assumption is that, absent the DST, prices for products sold via FBA and prices for the same products sold directly by Amazon would have followed parallel trends. One advantage of this strategy is that it compares prices for the same product within the same country and month. As we use country-specific regressions, common monthly shocks (e.g. lockdowns in response to the covid pandemic) affect both the treated and control observations and are absorbed by the time fixed effects.

A potential limitation of this strategy is that it may underestimate the true pass-through effect if Amazon also adjusts its own prices in response to the DST. Since third-party sellers on the marketplace face higher fees and may raise their prices accordingly, Amazon—acting as a direct competitor—might also increase its prices. If this occurs, the control group would no longer provide a fully unaffected counterfactual, leading to a downward bias in our estimate of the pass-through to consumers.

We therefore test whether Amazon's own prices respond to the DST by comparing prices for the same product across countries. The results, reported in Table B.3 in the appendix, show zero effects of the DST on Amazon prices: The coefficient is estimated at 0.001-0.002 and statistically insignificant in both a standard TWFE specification (with product-country and time fixed effects) and in a specification with product-time fixed effects. These findings suggest that Amazon's own prices do not systematically respond to the DST, making it unlikely that contamination of the control group materially affects our estimates.

We estimate the following regression equation separately for each country:

$$\ln(\text{Price})_{its} = \beta_1(FBA_{is} \times DST_t) + \delta_{is} + \delta_t + \epsilon_{its}. \quad (2)$$

The dependent variable, $\ln(\text{Price})_{its}$, is the natural logarithm of the price of product i at time t , sold by seller type s . The key explanatory variable, $FBA_{is} \times DST_t$, is an interaction term between an indicator variable FBA_{is} , which equals one if the seller of the product is a third-party seller using FBA, and DST_t , an indicator variable indicating whether the DST is in effect at time t . The coefficient β_1 captures the differential change in prices for FBA-sold products after DST implementation, relative to those sold directly by Amazon.

We include product-by-seller-type fixed effects, δ_{is} , to control for time-invariant product characteristics, and time fixed effects, δ_t , to account for common shocks affecting all products at a given time. Standard errors are clustered two-way by product

category and by month. Clustering at the product-category level allows for arbitrary correlation in residuals over time within a product category, which is the relevant unit of common supply-side and demand-side shocks in our data (e.g. category-level inventory disruptions, seasonal demand shifts, or shared cost structures). The time dimension additionally captures common macroeconomic and platform-wide shocks that affect all products within a given month, such as inflation or exchange-rate movements. Because each product-category cluster contains many individual products, clustering at the category rather than the product level is also strictly more conservative.

In Appendix A, we complement the DiD analysis with an instrumental variable (IV) approach. Specifically, we instrument the FBA fees with the introduction of the DST and estimate how the DST-induced changes in fees are passed through to consumer prices. This approach isolates the component of fee variation that is directly attributable to the tax and therefore does not rely on assumptions about Amazon's fee-setting behavior. It therefore provides an alternative estimate of pass-through that relies on cost shocks induced by the policy rather than on comparisons between seller types.

The instrument is relevant because the introduction of DSTs leads to measurable increases in FBA fees, as documented in the first part of our analysis. The identifying assumption is that the DST affects the prices set by third-party sellers only through its effect on the fees charged by Amazon. This assumption is plausible because the DST is levied on the platform rather than on the sellers themselves, making it unlikely that the tax directly affects sellers' pricing decisions through other channels.¹⁴

5 Data

5.1 Fee Data

To analyze the impact of DSTs on Amazon's fees, we use data on FBA fees. Amazon charges several types of fees to third-party sellers, most prominently referral fees and fulfillment fees. Referral fees are typically the most important fee, but no consistent data are available. We therefore focus on FBA fees, which are observable from publicly

¹⁴The exclusion restriction is not entirely straightforward in our setting. The FBA fee is only one component of the overall costs that sellers face when using the platform. The DST affects Amazon's platform revenue more broadly and may therefore influence several types of seller fees simultaneously (e.g. referral fees). To the extent that the DST induces changes in these additional fees, the IV estimates should be interpreted as capturing the pass-through of DST-induced increases in overall platform costs rather than the pass-through of FBA fees alone.

available rate cards. Amazon's FBA service includes storing sellers' products in its warehouses as well as handling packaging and shipping once an order is placed.

Figure 1: Fulfillment by Amazon (FBA) rate card

Calculating your fees

Simply add together your fulfilment and storage fees and any optional services you require to calculate your total FBA Fee.

1. Fulfilment Fee

A flat fee per unit, based on

- Dimensions
- Unit weight

2. Storage Fee (see page 4)

Charged per cubic metre per month or cubic foot per month

3. Optional Services Fee (see page 4)

Includes:

- Returns
- Disposal
- Labelling
- Bagging
- Bubble wrap
- Taping
- Opaque bagging

1. Fulfilment Fee per fulfilled unit¹

	Local fulfilment (domestic FBA, MCI, PanEU) within:									
	UK ²	DE ³ /PL/CZ	DE Only	FR ⁷	IT	ES	NL	SE	PL	
Small Envelope: Dimensions: ≤ 20 X 15 X 1 cm										
≤ 80g	£1.46	€ 1.73	€ 2.08	€ 2.44	€ 2.74	€ 2.26	€ 1.62	25.79 kr	4.48 zł	
Standard Envelope: Dimensions: ≤ 33 X 23 X 2.5 cm										
≤ 60g	£1.56	€ 1.88	€ 2.23	€ 2.60	€ 2.88	€ 2.51	€ 1.76	26.49 kr	4.53 zł	
≤ 210g	£1.66	€ 2.01	€ 2.36	€ 3.10	€ 3.03	€ 2.81	€ 1.93	27.64 kr	4.66 zł	
≤ 460g	£1.77	€ 2.16	€ 2.51	€ 3.48	€ 3.27	€ 3.02	€ 2.05	31.84 kr	4.91 zł	
Large Envelope: Dimensions: ≤ 33 X 23 X 5 cm										
≤ 960g	£2.21	€ 2.52	€ 2.87	€ 4.08	€ 3.53	€ 3.15	€ 2.48	33.09 kr	5.64 zł	
Standard Parcel: Dimensions: ≤ 45 X 34 X 26 cm										
≤ 150g	£2.19	€ 2.66	€ 3.01	€ 4.07	€ 3.62	€ 2.99	€ 2.53	37.14 kr	5.35 zł	
≤ 400g	£2.33	€ 2.95	€ 3.30	€ 4.86	€ 4.04	€ 3.50	€ 2.81	38.91 kr	5.66 zł	
≤ 900g	£2.50	€ 3.38	€ 3.73	€ 5.62	€ 4.66	€ 3.78	€ 3.26	39.45 kr	6.48 zł	

¹Additional per-unit fee of £0.10/€0.10/1.00kr/0.40zł for products that contain or are sold with lithium batteries and for dangerous goods (also costs associated with processing these products).

²This fee includes an additional charge of 2% for items sold on Amazon UK, to reflect Digital Service Tax introduced in UK w.e.f September 1st, 2020.

³This fee reflects a discount of £0.35/€0.35/3.50kr/1.50zł per fulfilled unit shipped when a seller participates in the Fulfilment Network Expansion (C-border fulfilment does not happen from a German Fulfilment Centre. Sellers who store inventory exclusively in Germany and do not participate in C-border fulfilment pay £0.35/€0.35/3.50kr/1.50zł per fulfilled unit. See FAQs on page 9 for more information).

⁴Units sold in Germany from Poland are charged the German domestic fee.

⁵Girth = Length (longest side) + 2x Width + 2x Height.

⁷This fee includes an additional charge of 3% for items sold on Amazon FR, to reflect Digital Service Tax introduced in FR w.e.f September 15, 2021.

Notes: Excerpts from Amazon's September 2021 EU FBA rate card. The panel on the left outlines how FBA fees are calculated; the panel on the right displays example fulfillment fees for standard-sized products. DST surcharges are mentioned for the UK and France, but not for Italy or Spain. *Source:* Amazon Seller Central.

Amazon publishes "rate cards" that specify the fees for each service separately, which allows us to calculate the total FBA fee for different product categories based on a product's weight and size (see Figure 1 for a snippet of such a rate card). FBA fees apply uniformly to all sellers for a given product category, size, and weight. The rate cards distinguish between local and cross-border shipping fees.¹⁵ In our analysis, we use only local shipments, as cross-border movements were significantly disrupted by Brexit and the COVID-19 pandemic during the period we study, and the tax treatment of cross-border services is not always clear. We focus in the main part of the analysis on standard-sized goods, as fees for these goods are more stable over time. Oversize goods account for only a small share of the sample (over 95% of the products in our pricing

¹⁵Shipping fees are determined based on the weight and dimensions of the packaged product, which classify items as either "Standard" or "Oversize." For example, products weighing 12 kg or more, or exceeding certain dimensional thresholds, are categorized as "Oversize," which incurs higher shipping fees.

data are standard-sized goods) and tend to be bulkier products with more volatile prices.

In addition to shipping fees, FBA fees include storage charges. Storage fees vary by product category,¹⁶ the volume stored, and the season. Rates are higher from October to December due to increased warehouse demand during peak shopping seasons.¹⁷

We calculate the FBA fee by summing the individual fees for storage and shipping. The rate cards provide these fees as pre-VAT amounts, and we use them as such in our analysis. Our analysis covers Germany, France, Spain, Italy, and the United Kingdom. We exclude data from the Netherlands, Sweden, and Poland because these marketplaces were included only from April 2020 onwards. For each country, we observe 180 distinct combinations of product type, package type, weight, and size.

We obtain the fee information starting in December 2017, when Amazon first released a consolidated rate card for all EU marketplaces, up to February 2022. In this period, Amazon published 14 rate cards. As fees did not change in between, we impute the data so that the final dataset has monthly observations. Amazon charges additional fees, such as a referral fee for using its marketplace. As we were unable to obtain information on these fees over time, our analysis focuses on FBA fees.

From April 2021 (September 2021) the rate cards include a note that the fees include an “additional charge” of 2% or 3% to cover the British (British and French) DSTs (see footnotes 2 and 5 in Figure 1). The Italian and Spanish DSTs are never mentioned on the rate cards during the sample period.

Table 2: Fee data: Descriptive statistics

	Mean	Weighted Mean	Min	p25	p50	p75	Max	Obs
France	6.77	5.18	2.83	4.39	6.99	8.65	12.49	2,346
Germany	5.66	4.39	2.83	3.74	5.95	7.16	9.59	2,499
Italy	6.31	4.83	3.12	4.07	6.27	8.04	11.48	2,244
Spain	5.56	4.30	2.57	3.83	5.65	7.08	9.76	2,397
UK	4.04	2.92	1.60	2.56	3.86	5.24	8.75	2,244

Notes: FBA fees for standard-sized products across weight, size, and packaging categories. Fees for France, Germany, Italy, and Spain are in EUR; UK fees are in GBP (exchange rate approx. 1.14 EUR/GBP). Weighted means use frequencies from the price dataset described in Section 5.2. The number of observations varies by country as some packaging types have zero weight. *Source:* Hand-collected Amazon rate cards, Dec 2017–Feb 2022. See Table B.1 for details.

¹⁶Initially, Amazon classified products as “media” or “non-media.” In April 2020, it introduced new categories “Clothing, Shoes and Bags,” and in June 2021, “Hazmat” (hazardous materials). We drop the fees for the “Hazmat” category.

¹⁷In addition, there are optional fees, e.g., for special packaging. We do not include these fees.

Table 2 provides descriptive statistics on the FBA fees. The fees vary between around EUR 1.80 and EUR 12.50. On average, the fees are lowest in the UK and highest in France.¹⁸

5.2 Price Data

For the second part of our analysis, we use product and pricing data from [Keepa.com](https://www.Keepa.com), a pricing consultancy. Keepa collects pricing information directly from Amazon, providing sellers with historical price trends to help optimize their pricing strategies. It tracks prices of products sold on Amazon (and also on other online marketplaces) across multiple countries.

The Keepa dataset covers all five countries in our study (United Kingdom, Germany, France, Italy, and Spain) and tracks several price categories for each product. We obtain data on the *FBA price*, which is the lowest price at which a product is offered by third-party sellers using the FBA service, and on the *Amazon price*, which is the price Amazon charges when it sells the product directly.¹⁹ All price data includes VAT but excludes shipping costs. The correlation between the two prices is high (0.96-0.97 in all countries).

In addition to pricing data, we also capture the number of sellers offering a product. This variable allows us to analyze whether changes in the number of suppliers affect the degree of tax pass-through. Furthermore, we obtain information on product identifiers, product category, packaging dimensions and weights.

Keepa collects data on a non-periodic basis, so products are tracked at irregular intervals and with varying frequency. We include only products with at least one observation in January 2019 or earlier, ensuring at least 12 months of pre-treatment data for each product before the first DST is implemented. Analogously to the analysis of FBA fees, we download data for the period December 2017 to February 2022 and restrict the sample to products that have both an Amazon price and an FBA price, with at least one observation before and after DST implementation in the respective marketplace. We use the Amazon Standard Identification Number (ASIN) and marketplace identifiers to define the cross-sectional unit.

We clean the raw price observations as follows. First, we winsorize prices at the 0.5% level within each country–marketplace to limit the influence of extreme outliers that

¹⁸The GBP to EUR exchange rate was around 1.13 EUR per GBP in 2018 and 1.17 EUR per GBP in 2022.

¹⁹Note that we only observe the lowest FBA price, but not which seller offered it. Thus, we cannot include seller fixed effects or compare the prices of the same seller across countries.

Table 3: Price data: Descriptive statistics

Full Sample								
Country	Price Type	p1	p10	p50	p90	p99	Mean	Obs
France	FBA Price	6.50	9.31	19.98	61.99	178.32	30.55	2 767 075
	Amazon Price	5.39	8.82	18.74	60.00	175.74	29.48	4 054 037
Germany	FBA Price	4.98	8.05	18.90	63.75	179.80	29.97	6 074 880
	Amazon Price	4.13	7.49	18.00	61.14	174.72	28.90	6 661 622
Italy	FBA Price	6.34	9.38	19.94	61.11	174.61	30.17	3 564 794
	Amazon Price	4.99	8.54	18.38	58.27	170.57	28.51	4 910 467
Spain	FBA Price	6.03	9.10	19.90	62.55	174.99	30.38	2 986 984
	Amazon Price	4.95	8.45	18.69	60.43	174.06	29.22	4 220 211
UK	FBA Price	3.32	5.27	12.07	37.27	114.48	18.57	4 955 066
	Amazon Price	2.99	5.22	11.74	36.85	113.31	18.17	6 372 401
Pre Treatment								
France	FBA Price	6.66	9.66	21.04	64.87	184.90	31.85	409 284
	Amazon Price	5.60	9.09	19.09	60.00	176.00	29.77	760 959
Italy	FBA Price	6.35	9.36	19.64	59.99	174.90	29.77	633 807
	Amazon Price	4.99	8.42	17.88	56.66	169.33	27.82	905 633
Spain	FBA Price	6.15	9.33	20.35	64.53	179.00	31.21	491 934
	Amazon Price	4.99	8.58	18.86	60.31	172.42	29.33	800 534
UK	FBA Price	3.26	5.14	12.13	38.07	115.20	18.72	842 966
	Amazon Price	3.00	5.19	11.68	36.39	112.26	18.05	1 202 221
Post Treatment								
France	FBA Price	6.47	9.49	20.63	63.63	177.70	31.27	465 029
	Amazon Price	5.51	9.10	19.08	60.88	177.12	29.98	736 823
Italy	FBA Price	6.50	9.77	20.57	63.30	179.00	31.15	612 504
	Amazon Price	5.17	8.82	18.53	58.02	169.99	28.57	907 266
Spain	FBA Price	5.96	9.10	20.16	64.34	175.00	30.98	465 017
	Amazon Price	4.95	8.55	18.95	61.00	174.00	29.51	763 103
UK	FBA Price	3.36	5.51	12.85	39.95	119.00	19.61	783 995
	Amazon Price	2.99	5.33	12.10	38.36	115.32	18.79	1 153 914

Notes: Descriptive statistics for the price data. Prices for France, Germany, Italy, and Spain are in EUR; UK prices are in GBP (exchange rate approx. 1.14 EUR/GBP). Values are based on monthly average prices per product. The full sample covers Dec 2017–Feb 2022. The pre- and post-treatment panels reflect monthly averages for the 9 months before and after DST implementation. *Data:* Keepa.com. See Table B.1 for details.

likely reflect recording errors or transient listing anomalies. We then aggregate the data to the product-month level by averaging all cleaned product-level observations within a month. This reduces noise from short-term price fluctuations or temporary discounts. Finally, we address missing observations using a local interpolation procedure: if some observations are missing between two observed months for the same product, we impute the missing value by linear interpolation of the adjacent observations.

Table 3 provides descriptive statistics for the price data used in our analysis. The table reports various percentiles as well as the mean of prices for two types of listings: FBA prices and Amazon’s own prices. The prices show some variation across countries and price percentiles. The median prices are close to EUR 20. Germany and, in particular, the UK exhibit somewhat lower prices than France, Spain, and Italy.

6 Results

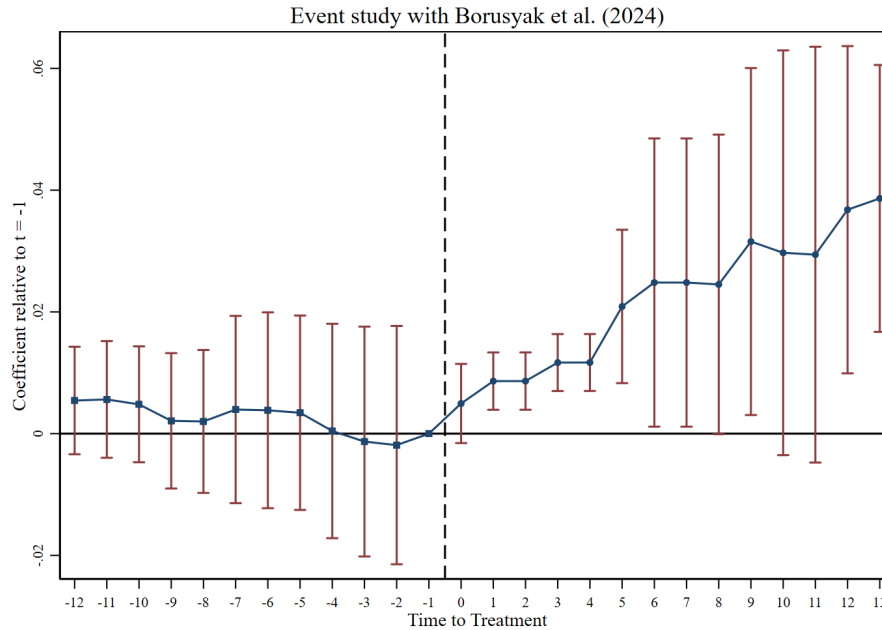
6.1 Marketplace Fees

We begin by analyzing how the introduction of DSTs affected the fees Amazon charges sellers for using its fulfillment services. As outlined in Section 4.1, we estimate equation (1) using a staggered DiD design, exploiting variation in DST implementation across countries. Our main outcome is the (log) FBA fee for a given product type, size, and weight category.

Figure 2 plots the dynamic treatment effects relative to the month of DST introduction, estimated using the imputation estimator proposed by [Borusyak et al. \(2024\)](#). Pre-treatment coefficients are small and statistically indistinguishable from zero, indicating parallel fee trends between treated and control countries prior to implementation. Following the introduction of the DST, fees increased immediately and remained persistently higher in the post-treatment period. The magnitude of the dynamic effects is economically meaningful and consistent with substantial pass-through of the tax to sellers. Event studies based on the estimators by [Callaway and Sant’Anna \(2021\)](#) and [De Chaisemartin and d’Haultfoeuille \(2024\)](#) yield very similar dynamic patterns (see Appendix Figure B.2).

To quantify the magnitude of the effects shown in Figure 2, Table 4 reports regression estimates across different specifications. In cols. (1) and (2), we use a traditional TWFE model. In column (1), the explanatory variable is the DST rate. The coefficient of 0.005 implies that a one percentage point increase in the DST raises FBA fees by 0.5%. Thus, about half of the DST is passed onto sellers.

Figure 2: Fee analysis: Event study



Notes: Event study estimates of the effect of DST implementation on FBA fees based on the imputation estimator by Borusyak et al. (2024). 95% confidence intervals. Standard errors clustered at the cross-sectional identifier. Data include only FBA fees for standard-sized products, weighted by their frequency in the price data. Monthly values are imputed by assuming fees remain constant between published rate cards. *Data:* Hand-collected Amazon rate cards, Dec 2017–Feb 2022. See Table B.1 for details.

Because the newer staggered DiD estimators are implemented with a binary treatment indicator, column (2) re-estimates the TWFE specification using a DST dummy instead of the continuous rate. The estimated coefficient of 0.020 implies that fees increase by 2.0% following DST implementation. Given statutory rates of 2% in the UK and 3% in France, Italy, and Spain, this magnitude is consistent with substantial pass-through of the tax into platform fees.

Columns (3) to (6) report estimates from modern staggered DiD estimators that account for heterogeneous treatment timing. Column (3) presents the imputation estimator by Borusyak et al. (2024), which corresponds to the event study in Figure 2. It implies an average increase in fees of 2.2% following DST implementation. Column (4) reports the estimator by Sun and Abraham (2021), yielding a larger effect of 5.0%. Column (5) shows the estimator by De Chaisemartin and d’Haultfoeuille (2024), which produces an estimate of 1.9%, close to the traditional TWFE result. Finally, column (6) reports results using the approach by Callaway and Sant’Anna (2021), with an overall estimate of 3.3%. Across all estimators, the evidence consistently points to economically meaningful pass-through of the DST into platform fees.

The differences in magnitudes across estimators reflect how they construct comparison groups and assign weights. The traditional TWFE specification in column (2) compares early- and late-treated countries as well as treated and never-treated units.²⁰ Column (3), the imputation estimator by [Borusyak et al. \(2024\)](#), re-weights group-time cells rather than cohorts and assigns relatively more weight to later post-treatment periods, where pass-through is somewhat larger, leading to a higher estimate. Column (4), the estimator by [Sun and Abraham \(2021\)](#), excludes already-treated units from the control group and aggregates cohort-specific effects across cohorts and event times. This gives greater influence to cohort-time cells with stronger positive effects. In our setting, the UK shows particularly strong effects in later post-treatment periods, which helps explain why this estimator yields a larger overall estimate. Column (5), the estimator by [De Chaisemartin and d'Haultfoeuille \(2024\)](#), relies on comparisons between switchers and not-yet-treated units within specific periods and places less weight on later treatment dynamics, producing an estimate close to TWFE. Finally, column (6), the [Callaway and Sant'Anna \(2021\)](#) estimator aggregates cohort-specific effects according to the share of treated units in each period, yielding an intermediate estimate. Overall, the range of estimates reflects alternative weighting schemes rather than disagreement about the direction of the effect.

The [Callaway and Sant'Anna \(2021\)](#) specification additionally allows us to estimate country-specific effects. We find close to full pass-through in France (2.5%) and Spain (2.4%) and more-than-full pass-through in the UK (6.8%). The effect for Italy is smaller (1.6%) and statistically insignificant.

One potential explanation for the muted response in Italy is that Amazon may have been more cautious in adjusting fees in a market where its position was still developing. Compared to the other countries in our sample, Amazon had a relatively small market share in Italy in 2020 (17.8%, compared to 22.4% in Spain and 36.2% in Germany).²¹ Amazon also entered the Italian market relatively late: its Italian site ([amazon.it](#)) launched only in 2010, much later than in the UK and France. Italy was then the first country to collect a DST. In line with this interpretation, Appendix Table B.5 shows that Italy is the only country where the number of third-party sellers increased following DST implementation. Raising marketplace fees in such a phase

²⁰As shown in Table B.2, comparisons between timing groups account for 53% of the identifying variation but contribute a near-zero estimate. The identifying variation from treated-versus-never-treated comparisons accounts for the remaining 47% and yields a substantially larger estimate. This pattern suggests that the TWFE estimate is attenuated by comparisons across treatment-timing groups.

²¹See [Statista Market Insights \(2025\)](#); [About Amazon Europe \(2021a,b\)](#); [Amazon.com, Inc. \(2021\)](#) for the market share information.

could discourage entry or expansion by sellers. Amazon may therefore have chosen to keep its fees largely stable—effectively absorbing most of the DST—to continue growing the platform.

Table 4: Fee analysis: Main regression results

Estimator	TWFE		BJS	SA	DCDH	CS
Expl. Var.	DST Rate %			Dummy		
	(1)	(2)	(3)	(4)	(5)	(6)
DST	0.005* (0.003)	0.020** (0.009)	0.022** (0.009)	0.050*** (0.002)	0.019*** (0.003)	0.033*** (0.011)
France						0.025** (0.011)
Italy						0.016 (0.015)
Spain						0.024** (0.009)
UK						0.068*** (0.022)
Controls	✓	✓				
Cross-section FE	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓
R^2	0.985	0.985		0.991		
Observations	11,730	11,730	10,760	11,730	11,730	11,730

Notes: Estimated effects of digital services tax (DST) implementation on Fulfillment by Amazon (FBA) fees across different estimators. Columns (1) and (2) use the traditional two-way fixed effects (TWFE) estimator, with column (1) based on the DST rate and column (2) on a DST dummy. Column (3) uses the imputation estimator by [Borusyak et al. \(2024\)](#), column (4) the estimator by [Sun and Abraham \(2021\)](#), column (5) the estimator by [De Chaisemartin and d'Haultfoeuille \(2024\)](#), and column (6) the estimator by [Callaway and Sant'Anna \(2021\)](#), which also reports country-specific effects. Only standard-sized products are included. Observations are weighted by frequency in the price data. Monthly values are imputed by assuming constant fees between published rate cards. Standard errors clustered at the cross-sectional identifier. ***, **, * indicate significance at the 1%, 5%, and 10% levels. *Data:* Hand-collected Amazon rate cards, Dec 2017–Feb 2022. See Table B.1 for details.

To better understand the cross-country variation underlying the average effects, and the differences across estimators, Figure B.3 in the Appendix presents country-specific event studies (for a period of 13 months before/after DST introduction). In the UK and France, fees increase in the month of DST implementation, indicating immediate pass-through to sellers. In Spain, fees rise only about six months after implementation. This delay coincides with the first publication of a new rate card after the introduction of the DST, suggesting that the DST was incorporated when fees were updated. In the UK, effect sizes are similar to those in the other countries, indicating that the larger coefficients in Table 4 were driven by later periods. Italy exhibits no statistically

significant adjustment in fees following DST introduction, consistent with the smaller average effect reported in Table 4.

We cluster standard errors at the cross-sectional identifier. This level matches the unit at which we construct the weights and at which the staggered DiD estimators in columns (3)–(6) operate. Clustering at the country level is infeasible: with only one never-treated country, the standard asymptotic approximations break down. Appendix Table B.4 reports results when we instead cluster at the country $\times$ product-type level. The staggered DiD estimates in columns (3)–(6) are essentially unchanged; only the weighted TWFE estimates in columns (1) and (2) lose statistical significance. This reflects a mechanical interaction: the weights are computed at the cross-sectional level, so clustering above that level inflates standard errors.²²

We next examine the robustness of the baseline specification based on the continuous DST rate (Table 4, column 1). The results are reported in Table 5. In Panel A, column (1), we re-estimate the regression without weights, and in column (2), we additionally use the raw, unbalanced data with irregular time intervals. Both specifications yield slightly larger coefficients, implying pass-through close to 100%. This pattern indicates that fee increases were somewhat stronger in less frequently used product categories. Because the baseline specification weights observations by their frequency in the price dataset, the overall average effect is moderated by the smaller adjustments observed in the most commonly used fee categories.

Panel B explores heterogeneity across product characteristics. Column (1) uses data on fees for oversized goods and yields similar effects. Columns (2) and (3) split standardized products into media and non-media categories. The estimated coefficients are nearly identical across categories, indicating that Amazon did not differentially adjust fees across product types.

²²As a further robustness check, we re-estimate columns (1) and (2) with Driscoll–Kraay standard errors using a Bartlett kernel and a Newey–West-selected bandwidth (Driscoll and Kraay, 1998). The point estimates are unchanged; the dummy specification in column (2) remains significant at the 10% level, and the rate specification in column (1) is marginally insignificant. All imputed specifications in Table 5 (columns 1, 3, 4, and 5 of the heterogeneity table) remain significant under Driscoll–Kraay; the non-imputed specification cannot be estimated this way because the estimator requires a sufficiently long time series.

Table 5: Fee analysis: Robustness and heterogeneity

	Panel A: Robustness		Panel B: Heterogeneity		
Product size	Standard		Oversize	Standard	
Product type	All			Media	Non-Media
	(1)	(2)	(1)	(2)	(3)
DST Rate	0.010*** (0.002)	0.008*** (0.002)	0.013*** (0.002)	0.010*** (0.003)	0.010*** (0.002)
Controls	✓	✓	✓	✓	✓
Cross-Section FE	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓
Imputed	✓	–	✓	✓	✓
R^2	0.987	0.988	0.985	0.987	0.986
Observations	15,300	3,600	19,125	5,100	10,200

Notes: Robustness and heterogeneity tests for the effect of DSTs on FBA fees. Panel A tests robustness to weighting and imputation. Column (1) removes weights; column (2) additionally only uses original (non-imputed) observations. Panel B explores heterogeneity by product size and category (again un-weighted): column (1) includes only oversize goods, columns (2) and (3) split the sample into media and non-media products. All regressions include inflation, the market share of the digital economy, and a COVID-19 policy stringency index as controls. Only standard-sized products are included, except in Panel B, column (1). Monthly values are imputed by assuming constant fees between published rate cards, except in Panel A, column (2). Standard errors clustered at the cross-sectional identifier. ***, **, * indicate significance at the 1%, 5%, and 10% levels. *Data:* Hand-collected Amazon rate cards, Dec 2017–Feb 2022. See Table B.1 for details.

6.2 Prices

Are the observed increases in fees passed on to consumers via higher prices? To analyze this question, we compare price changes for identical products sold by two groups of sellers on Amazon: third-party sellers using FBA (treated group) and Amazon as a direct seller (control group). Since DSTs apply only to marketplace transactions, this allows us to estimate the direct effect of DSTs on prices. We focus on the medium term, using monthly data for nine months before and after the introduction of the DST. We estimate eq. (2) separately for each country.

Table 6 presents the results. The coefficient on $FBA_{is} \times DST_t$ captures the differential change in prices for products sold by third-party sellers using FBA relative to the same products sold directly by Amazon after the introduction of the DST. We include product-by-seller-type and month fixed effects. Across most countries, the estimated coefficients are positive and statistically significant. Prices for products sold by FBA sellers increase by about 2% following DST implementation in France, Spain, and the United Kingdom. By contrast, the estimated effect in Italy is very small and

statistically insignificant. This pattern mirrors the results for Amazon's fees reported in Section 6.1, where we also find little evidence of fee increases in Italy. In Section 7, we will calculate the implied incidence on consumers for each country.

Table 6: Price analysis: DiD results

	(1) France	(2) Italy	(3) Spain	(4) United Kingdom
$FBA_{is} \times DST_t$	0.020* (0.011)	0.001 (0.007)	0.021** (0.008)	0.022** (0.009)
Product-by-seller-type FE	✓	✓	✓	✓
Time FE	✓	✓	✓	✓
R ²	0.97	0.97	0.97	0.96
Observations	2,372,095	3,059,210	2,520,588	3,983,096

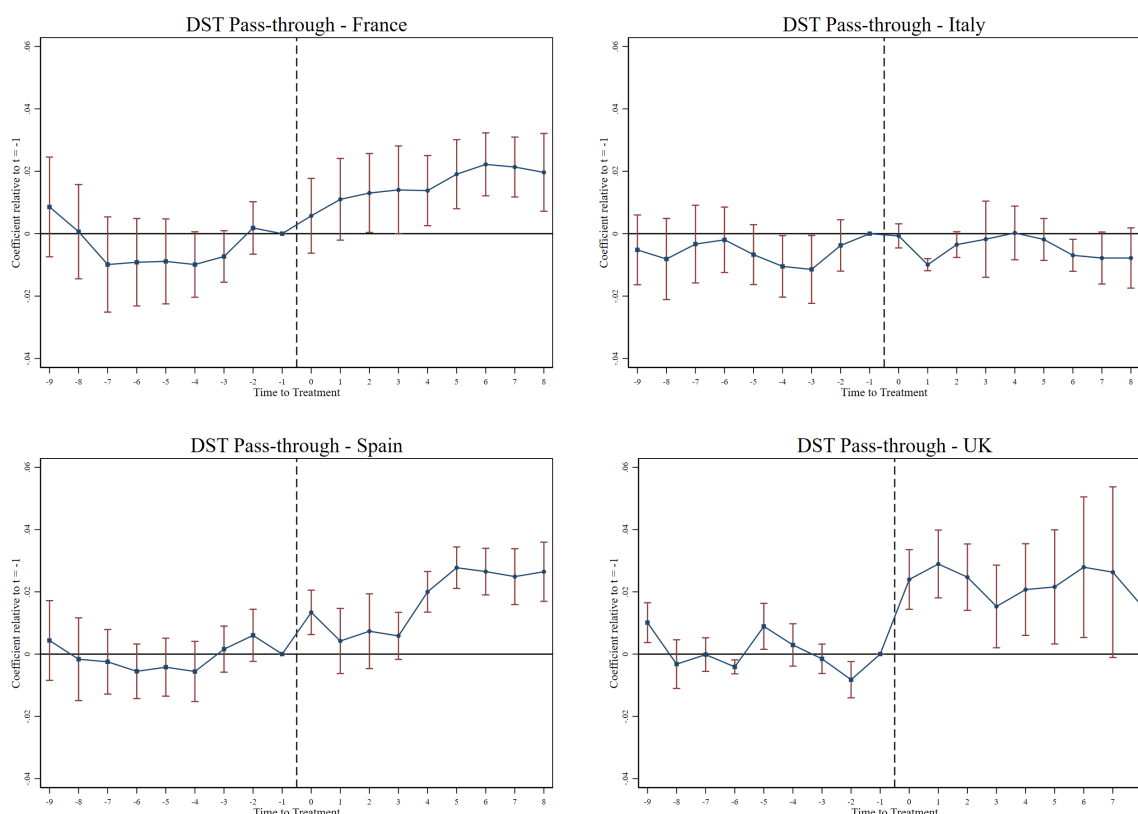
Notes: Estimated effects of DST implementation on consumer prices, using a DiD specification. The outcome is the logarithm of the price for products sold by third-party sellers using FBA, compared to prices of identical products sold by Amazon directly. All columns include product-by-seller-type and time fixed effects. Only standard-sized products are included. Data for nine months before and after DST introduction. Standard errors are clustered two-way by product category and by month. ***, **, * indicate significance at the 1%, 5%, and 10% levels. *Data:* Keepa.com. See Table B.1 for details.

Figure 3 shows the country-level event studies, comparing FBA prices to Amazon's own prices, again separately for each of the countries that have introduced a DST. We see an increase in prices after the introduction of the DST in France, Spain, and the UK, but not in Italy. In Spain, this increase occurs only with a delay, mirroring the delayed rise in Amazon's fees documented above and suggesting that prices responded once sellers faced higher platform fees. All pre-treatment coefficients in France, Spain, and the UK are statistically indistinguishable from zero. These findings are consistent with our main results indicating that DSTs meaningfully increase marketplace sellers' prices.

To ensure that our results are not affected by interpolating missing observations in the price data, we re-estimate the results with only original observations. The results (see Table B.6 in the appendix) are very similar.

One limitation of our DiD approach is that Amazon may also adjust its own prices in response to the DST, as it competes with third-party sellers on its platform. If Amazon raises its prices, our control group would be partially affected, leading to a potential underestimation of the true pass-through. To assess this concern, Appendix Table B.3 examines whether the DST affected prices of products sold directly by Amazon. Column (1) estimates a TWFE specification with product-country and time fixed effects, while column (2) compares prices of the same product across countries within

Figure 3: Price analysis: Event studies by country



Notes: Event study estimates of the effect of DST implementation on consumer prices, comparing products sold by third-party sellers using FBA to the same products sold directly by Amazon. Each panel shows log prices centered on the month of DST introduction (vertical line), with a window of 9 months before and after. 95% confidence intervals. Standard errors clustered two-way by product category and by month. *Data:* See Table B.1.

the same month using product–time fixed effects. Both specifications yield statistically insignificant coefficients close to zero, suggesting that Amazon did not systematically adjust its own prices following DST implementation.

Nevertheless, the DiD specification identifies price effects by comparing seller types and therefore relies on the assumption that Amazon’s own prices provide a valid counterfactual. To estimate the relationship between Amazon’s fees and consumer prices more directly, we additionally implement an IV approach, where we use the DST as an instrument for Amazon’s fulfillment fees. Appendix A provides details on the specification and estimation.

The IV estimates in Table A.1 confirm substantial pass-through. The first-stage F-statistics are well above conventional thresholds in all specifications. The second-stage coefficients range from about 3.4 to 6.8, with larger estimates when only FBA prices are used (columns 1–2) and smaller ones when Amazon’s own prices serve as

an additional within-product control (columns 3–4). Part of this decline is mechanical: Amazon’s own sales have zero FBA fees by construction, so including them pulls the time fixed effects in the first stage toward zero and mechanically increases the first-stage coefficient, which in turn lowers the second-stage IV estimate.

All IV coefficients exceed one, implying that prices rise by more than the observed increase in FBA fees. Two factors contribute to this. First, the DST also affects referral fees, which are proportional to the sales price and which we do not observe in our data. Since referral fees account for a substantial share of total seller costs, these large estimates reflect sellers passing through both FBA and referral fee hikes, rather than an over-shifting of FBA fees alone. Second, the coefficients are consistent with some degree of overshifting, which we will discuss further in Section 7.

6.3 Effect Heterogeneity

The extent of pass-through may vary across products depending on their pricing environment. As discussed in Section 3, in classical models the tax incidence depends on market structure and elasticities. In addition, the DST increases Amazon’s fees in absolute terms, while our regressions estimate percentage changes in prices. As a result, a given fee increase represents a larger cost shock relative to the price of cheaper products. To explore these mechanisms, we examine heterogeneity along three dimensions: First, we split products by their pre-treatment price level. Second, we split products by the volatility of pre-treatment prices, which captures how frequently prices adjust. Finally, we examine heterogeneity by the number of third-party sellers offering a product, which provides a proxy for the competitive environment on the marketplace.

Figure 4 shows the event-study estimates separately for products with pre-treatment prices below and above the median. In France, Spain, and the UK, price responses are clearly larger for cheaper goods. Prices for products below the median increase significantly following DST implementation, whereas prices for more expensive products rise only slightly, and the increases remain statistically insignificant. Italy is the only country where we do not observe this pattern, consistent with the overall nonresponse.

This pattern is consistent with the mechanisms discussed above. First, the DST increases Amazon’s fees in absolute terms, while our regressions estimate percentage changes in prices. A given increase in fees therefore represents a larger cost shock relative to the price of cheaper products, which mechanically leads to stronger price

responses in this group.²³ Second, pricing frictions may reinforce this pattern. Retail prices often cluster at salient thresholds such as prices ending in “.99”, so that small cost increases may lead sellers to move to the next price point rather than adjusting prices marginally (Conlon and Rao, 2020). Both mechanisms imply stronger pass-through for lower-priced goods.

Figure 5 shows the results when splitting products by the volatility of their pre-treatment prices (measured by the standard deviation of pre-DST FBA prices). In Spain and the United Kingdom, we find clear price increases for products with below-median volatility, while prices for products with more volatile price histories do not respond significantly to the DST. In France, the differences between the two groups are smaller, and in Italy we again find no significant effects.

One possible explanation is that high price volatility reflects the use of automated repricing algorithms. In online retail markets, sellers increasingly rely on software that frequently updates prices and reacts automatically to competitors’ price changes (Brown and MacKay, 2023). If prices are largely determined by such algorithms, pricing decisions may respond more strongly to competitors’ prices than to cost changes. By contrast, products with more stable prices may involve less automated pricing and more deliberate adjustments, making it more likely that increases in platform fees are reflected in retail prices.

Finally, we examine heterogeneity by the number of sellers offering a product. Figure 6 shows the results when splitting products by the median number of third-party sellers. In France and Spain, price responses tend to be somewhat larger for products with fewer suppliers. In the United Kingdom, by contrast, the price responses look similar across both groups—if anything, the response is slightly larger for products with more suppliers. In Italy, we again find no significant effects.

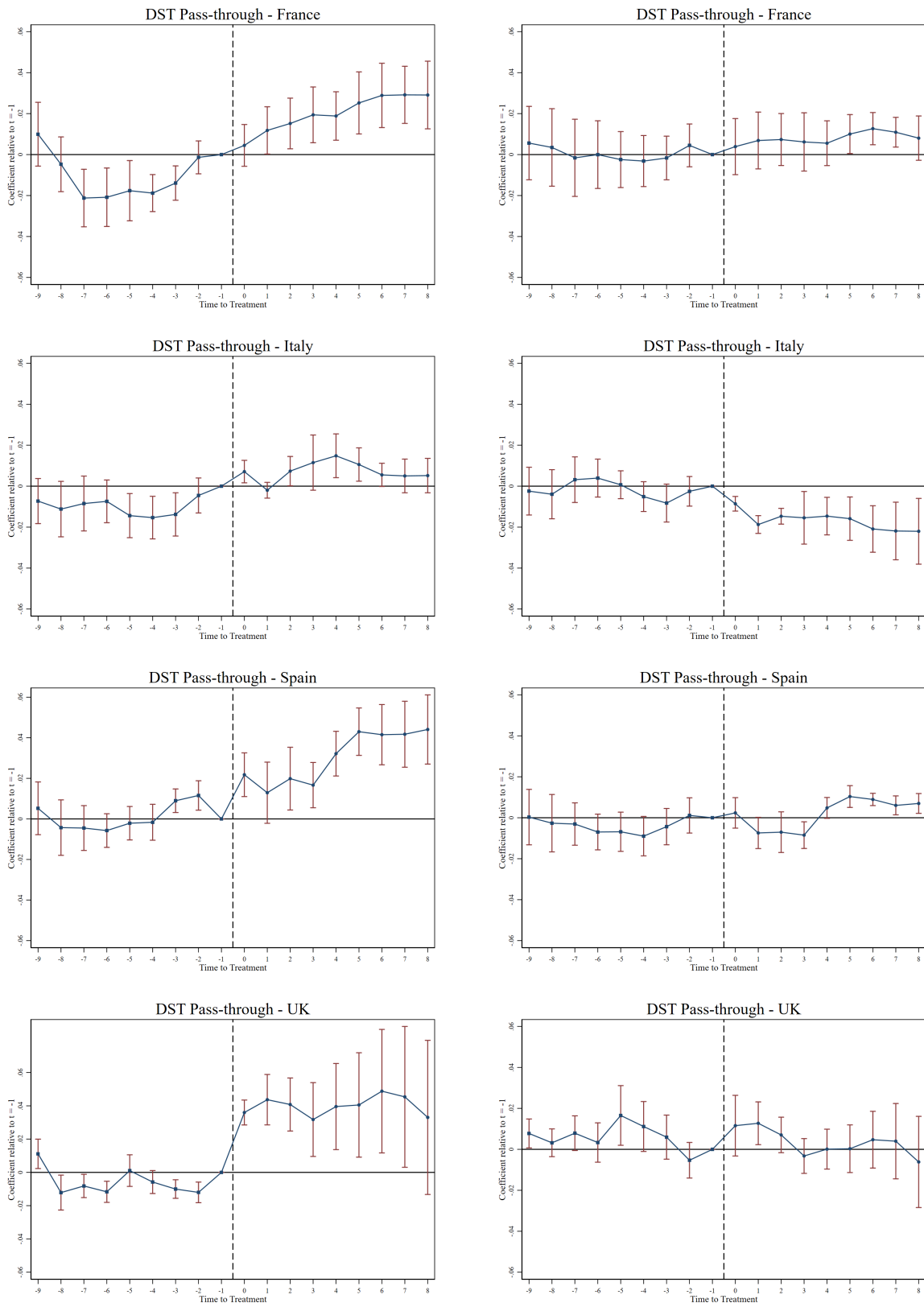
The number of sellers may capture two opposing forces. On the one hand, a larger number of sellers likely indicates stronger competitive pressure. In standard models, this can strengthen pass-through, as sellers have less scope to absorb cost increases in their margins. On the other hand, products with many sellers may be more likely to be priced algorithmically, so that prices react more strongly to competitors’ prices than to own cost shocks. This can weaken the direct pass-through of higher platform fees.

²³As an additional check, we split products by their pre-treatment FBA fees. The results show somewhat stronger price responses for products with above-median FBA fees in Spain, France, and the United Kingdom, while no clear pattern emerges for Italy. This pattern is consistent with the mechanism discussed in the main text: the relevant cost shock is the fee increase relative to the product price. A higher baseline FBA fee raises the numerator of this ratio, just as a lower product price reduces its denominator, so both margins imply a larger proportional shock and potentially stronger price adjustment.

Figure 4: Price analysis: Heterogeneity by pre-treatment price

(a) Below Median

(b) Above Median

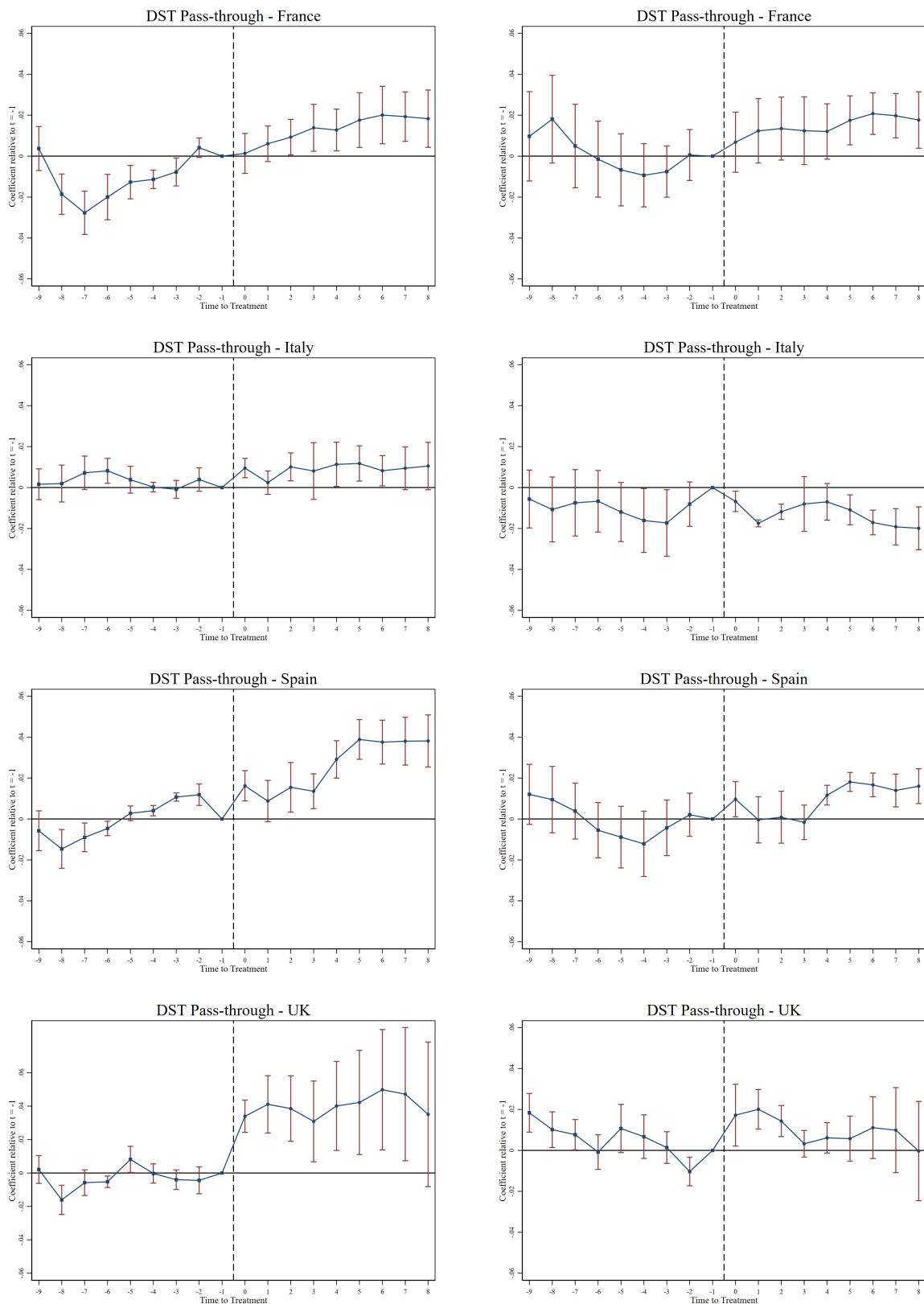


Notes: Event-study estimates split by the median of the average pre-treatment FBA price. In each row, the left panel shows products below the median and the right panel shows products above the median. 95% confidence intervals. Standard errors clustered two-way by product category and by month. *Data:* See Table B.1.

Figure 5: Price analysis: Heterogeneity by pre-treatment price volatility

(a) Below Median

(b) Above Median

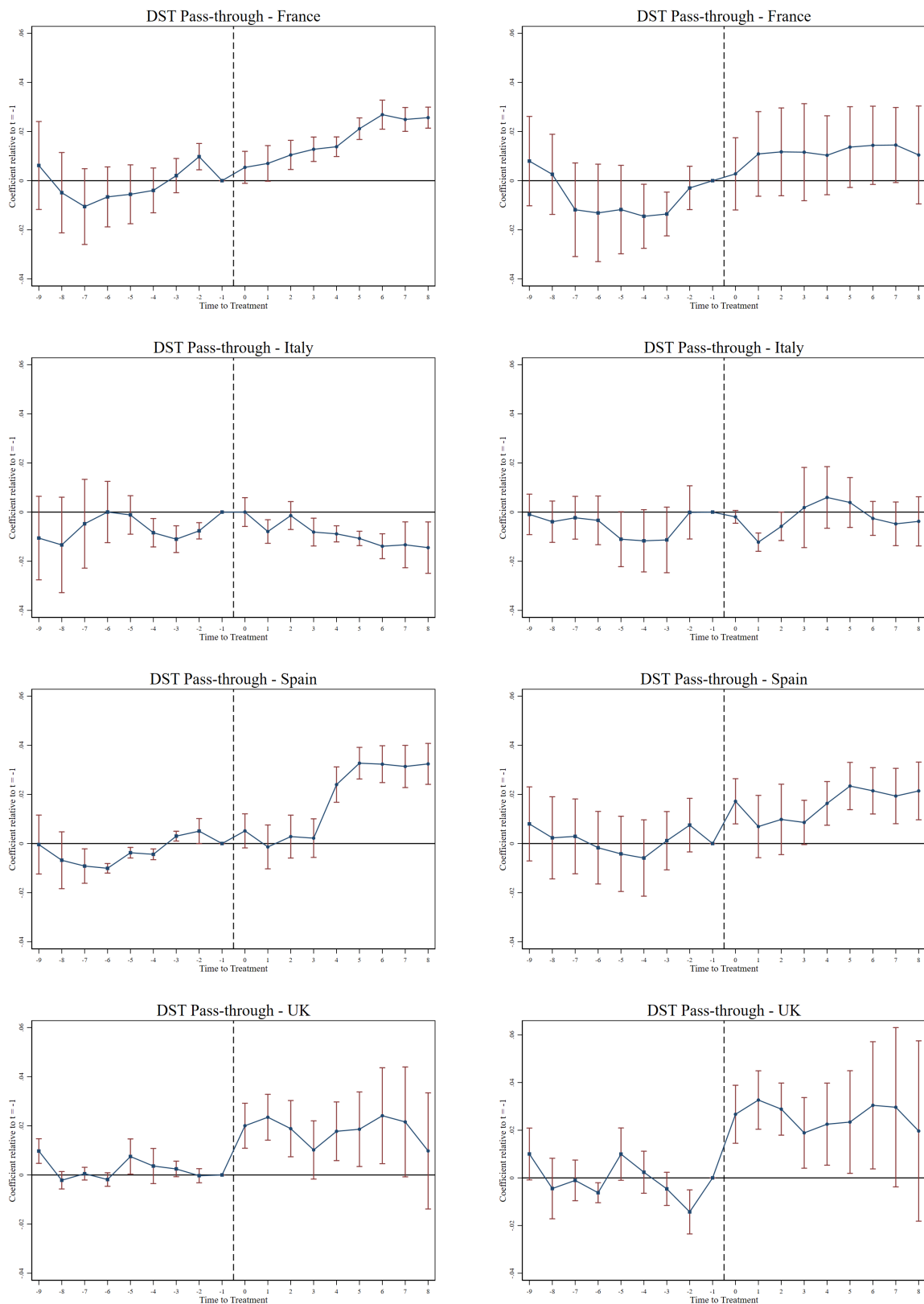


Notes: Event-study estimates split by the median of pre-treatment price volatility, measured as the average standard deviation of pre-treatment FBA prices. In each row, the left panel shows products below the median and the right panel shows products above the median. 95% confidence intervals. Standard errors clustered two-way by product category and by month. *Data:* See Table B.1.

Figure 6: Price analysis: Heterogeneity by number of suppliers

(a) Below Median

(b) Above Median



Notes: Event-study estimates split by the median number of suppliers. In each row, the left panel shows products below the median and the right panel shows products above the median. 95% confidence intervals. Standard errors clustered two-way by product category and by month. Data: See Table B.1.

The results suggest that these two forces may differ in importance across countries: in France and Spain, the algorithmic-pricing channel may be more relevant, whereas in the United Kingdom the competition channel appears to dominate.

7 Incidence Calculation

We now use our estimates to assess the incidence of the DST. The goal of this section is to translate the estimated effects on fees and prices into economically meaningful measures of who ultimately bears the tax. We first look at the two margins (pass-through from Amazon to third-party sellers through fulfillment fees, and pass-through from sellers to consumers through retail prices) separately. We then relate the resulting price changes to the total DST liability generated by a transaction to calculate the ultimate share borne by consumers.

To do so, we need to make some additional assumptions. The DST applies to all marketplace fees. As we do not observe referral fees, we assume that it equals 15% of the net-of-VAT sales price when calculating total DST liability.²⁴ We also assume that referral fees rise by the same proportion as FBA fees in response to the DST.

Table 7 then summarizes how the DST is transmitted along the supply chain. Column (1) reports the implied increase in fulfillment fees, while column (2) relates this increase to the DST applied to fulfillment services, providing a measure of pass-through from Amazon to sellers. Column (3) reports the implied change in consumer prices net of VAT, and column (4) relates this change to the increase in fees, capturing pass-through from sellers to consumers. Seller-to-consumer pass-through depends on how referral fees respond to the DST. In the table, we assume that they respond by the same proportion as FBA fees.²⁵ Finally, column (5) relates the price change to the total DST liability generated by a transaction, including both fulfillment and referral fees.

The estimates indicate substantial pass-through along both margins, with clear differences across countries. In France and Spain, Amazon passes on roughly 80% of the DST applied to fulfillment fees to third-party sellers, while absorbing the remaining share. Sellers, in turn, more than fully pass these higher costs on to consumers, with

²⁴In some cases, additional fees apply, such as closing fees for media items, which we do not include in the calculation.

²⁵For Spain, for example, the implied seller-to-consumer pass-through ranges from 1.98 (if referral fees increase by the full DST of 3%) to 3.42 (if referral fees remain constant). Under the intermediate assumption that referral fees rise proportionally with FBA fees, the implied pass-through is 2.16.

Table 7: Incidence calculation

	Fee increase (EUR/GBP)	Amazon-to-seller pass-through	Price change (EUR/GBP)	Seller-to-consumer pass-through	Total consumer incidence
France	0.13 [0.02, 0.25]	0.84 [0.11, 1.59]	0.35 [0.21, 0.49]	1.80	1.51 [0.92, 2.11]
Italy	0.08 [-0.06, 0.22]	0.54 [-0.44, 1.55]	0.02 [-0.05, 0.08]	0.14	0.07 [-0.22, 0.37]
Spain	0.10 [0.04, 0.17]	0.81 [0.28, 1.35]	0.36 [0.26, 0.46]	2.16	1.74 [1.25, 2.24]
UK	0.21 [0.07, 0.34]	3.52 [1.26, 5.88]	0.22 [0.16, 0.29]	0.74	2.53 [1.85, 3.22]

Notes: This table summarizes the transmission of DSTs to seller fees and consumer prices. Column (1) reports the implied increase in the weighted-mean FBA fee (from Table 2) based on the estimated country-specific fee effects from column (6) of Table 4. Column (2) relates this increase to the DST applied to fulfillment fees, providing a measure of Amazon-to-seller pass-through. Column (3) reports the implied change in consumer prices net of VAT, evaluated at the pre-treatment median price in each country (from Table 3) based on the estimates from Table 6. Column (4) reports seller-to-consumer pass-through, defined as the ratio of the price change to the total increase in marketplace fees, including both fulfillment and referral fees. We assume a referral fee equal to 15% of the net-of-VAT sales price and that referral fees increase proportionally with FBA fees. Column (5) relates the price change to the total DST liability generated by a transaction, again including both fulfillment and referral fees. Confidence intervals in columns (1) and (3) are based on the estimated fee and price effects, respectively; columns (2) and (5) scale these intervals accordingly. Confidence intervals are not reported for column (4), as this measure combines estimated fee and price effects and requires an additional assumption on referral-fee adjustments. *Data:* See Table B.1.

price increases that are roughly twice as large as the increase in fees. As a result, consumer prices rise by around 1.5 to 1.7 euros for each euro of DST liability generated by a transaction.

In the UK, Amazon more than fully passes the DST on to sellers through higher fulfillment fees, with pass-through exceeding 300%. Sellers then pass about 74% of these increases on to consumers, leading to an overall incidence on consumers of about 2.5. This pattern implies that Amazon does not bear the tax burden and instead over-shifts it to sellers, who in turn transmit it to consumers. In Italy, by contrast, the estimated increase in fulfillment fees is small and statistically imprecise, and the price response is close to zero. This suggests that Amazon absorbs most of the DST and that there is little pass-through to consumers.

The estimated incidence on consumers implies more-than-full pass-through to consumers in France and Spain. Such overshifting is well documented in the literature (Kenkel, 2005; Bonnet and Réquillart, 2013; Kopczuk et al., 2016; Hindriks and Serse, 2019). Why does it occur? In markets with imperfect competition, firms may adjust markups in response to cost increases, so that consumer prices rise by more than the

tax itself. Then, the extent of tax pass-through depends not only on demand and supply elasticities but also on the curvature of demand and the strategic interaction among firms (Seade, 1985; Besley, 1989).

In addition, pricing frictions may contribute to overshifting. Retail prices often cluster at psychological thresholds such as prices ending in “.99”, so that small cost increases can lead firms to move prices to the next salient price point rather than adjusting them marginally (Conlon and Rao, 2020). In the context of Amazon’s marketplace, where sellers frequently rely on algorithmic pricing and price points designed to maximize consumer attention, such pricing frictions may reinforce the tendency toward more-than-full pass-through.

8 Conclusion

This paper investigates the economic incidence of DSTs through the case of Amazon, the largest online retailer. Using detailed data on platform fees and consumer prices, we find that in most countries, Amazon passes the DST largely onto sellers via increased fees. These higher costs are largely shifted to consumers, though the extent of pass-through varies across countries. Notably, we observe full pass-through or even overshifting in the UK, France, and Spain, but almost no pass-through in Italy.

Our findings have important implications for current policy debates. If the objective of the DST is to tax large digital corporations, our results suggest that this goal is largely not achieved in practice. In the case of Amazon, the tax is almost fully passed on to third-party sellers through higher platform fees and subsequently to consumers through higher retail prices. The economic burden therefore falls predominantly on domestic market participants rather than on the targeted multinational firm. From an equity and efficiency perspective, this raises concerns about the effectiveness of revenue-based digital taxation as a tool to tax profits in the digital economy.

However, if the policy objective is instead to reduce the competitive advantage of online platforms relative to brick-and-mortar retailers, the mechanism we document may operate in the intended direction. By increasing prices in the digital marketplace, the DST narrows the relative price gap between online and offline commerce. In this sense, the tax functions less as a profit tax on multinational firms and more as a consumption tax on digital transactions. This interpretation is particularly relevant in the context of recent transatlantic trade tensions, where a coordinated EU-wide DST has been discussed as a potential response to U.S. tariffs. Economically, such a measure would resemble a tariff on digital services: it would raise the cost of cross-border digital

intermediation and would likely be largely borne by domestic consumers and sellers. As with conventional tariffs, policymakers must therefore weigh its strategic trade value against the domestic welfare costs it imposes.

Our analysis is not without limitations. Our focus on Amazon likely does not fully capture the dynamics on other digital marketplaces or on other business models that fall under the scope of the DST. Additionally, we cannot fully explain the observed variation in pass-through across countries. Future research could explore the broader impacts of DSTs on market competition, innovation, and platform strategies, as well as their interaction with other tax instruments.

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A Instrumental Variable Analysis

We instrument the FBA fee with the euro amount of the DST due. The structural equation is

$$\text{Price}_{ict} = \beta_1 \text{FBA fee}_{ict} + \delta_i + \delta_t + \varepsilon_{ict}, \quad (\text{A.1})$$

where Price_{ict} is the price of product i in country c at time t , δ_i denotes product fixed effects, and δ_t denotes time fixed effects. Because a product's dimensions and weight do not change, product fixed effects absorb fee-category fixed effects. ε_{ict} is the error term. The coefficient β_1 captures how much of the fee is passed on to consumers; a value of 1 implies full pass-through.

The first-stage equation is

$$\text{FBA fee}_{ict} = \pi_1 Z_{ic} + \delta_i + \delta_t + u_{ict}. \quad (\text{A.2})$$

Both stages include the same fixed effects and controls. In some specifications, we replace δ_t with product group–time fixed effects. The instrument Z_{ic} is the euro amount of the DST due for a given product:

$$Z_{ic} = \tau_c \times \overline{\text{FBA fee}_{ic}}, \quad (\text{A.3})$$

where τ_c is the DST rate in country c in decimal form (e.g., 0.03 for a 3% tax) and $\overline{\text{FBA fee}_{ic}}$ is the pre-treatment FBA fee for product i in country c . The instrument is computed once and held constant across all post-treatment periods; it equals zero before treatment and in countries without a DST. Because Z_{ic} is denominated in euros, $\pi_1 = 1$ in the first stage corresponds to full pass-through of the tax to fees. By fixing the fee at its pre-treatment value, we avoid any mechanical relationship between the instrument and contemporaneous fee changes, but we also lose within-product fee variation over time. The first-stage relationship is therefore estimated less precisely than the fee analysis in the main text, which exploits the full time series of fee changes.

We estimate the model on two samples. Columns (1)–(2) use only FBA prices, with Germany as control. Columns (3)–(4) additionally include Amazon's own prices as a within-product control group, following the same logic as the DiD analysis in Section 6.2; product fixed effects in these columns correspond to product-by-seller-type fixed effects.

Table A.1 presents the results. The first-stage coefficients are positive and significant in all specifications, with F-statistics between 15 and 27. The coefficients for the first stage range from about 1 to 2.2, indicating that fees rise by at least as much as

the DST due. These coefficients exceed those in the fee analysis in the main text, likely because the instrument is fixed at the pre-treatment fee level and therefore does not capture subsequent fee adjustments unrelated to the DST.

Table A.1: Price analysis: IV results

	FBA prices only		FBA and Amazon prices	
	(1)	(2)	(3)	(4)
FBA Fee	6.844*** (1.499)	5.953*** (1.218)	3.799*** (0.623)	3.391*** (0.572)
1 st stage coeff.	0.877***	1.504***	1.967***	2.189***
1 st stage F-stat.	15.05	15.96	25.58	26.65
Controls	✓	✓	✓	✓
Product FE	✓	✓	✓	✓
Time FE	✓		✓	
Prod.group-time FE		✓		✓
Obs.	20,347,993	20,347,535	46,566,160	46,565,916

Notes: Instrumental variable estimates of the effect of FBA fees on consumer prices. The FBA fee is instrumented with $Z_{ic} = \tau_c \times \overline{\text{FBA fee}}_{ic}$ (see text). Columns (1)–(2) use only FBA prices with Germany as the control group. Columns (3)–(4) additionally include prices of products sold directly by Amazon as a within-product control group; product fixed effects in these columns correspond to product-by-seller-type fixed effects. All specifications control for inflation, the market share of the digital economy, and a COVID-19 policy stringency index. The dependent variable is the price, aggregated to monthly values. Standard errors are two-way clustered by product and by country–month. ***, **, * indicate significance at the 1%, 5%, and 10% levels. *Data:* See Table B.1.

The second-stage estimates indicate substantial price responses. When using only FBA prices (columns 1–2), a one-euro increase in FBA fees raises prices by about EUR 6–7. With the within-product control group (columns 3–4), the estimates are smaller at about EUR 3.4–3.8. All coefficients exceed one, consistent with more-than-full pass-through. This is expected: the DST also affects referral fees, which are proportional to the sales price. Using mean prices and fees in our sample, referral fees account for roughly one-third of total marketplace revenues, so an IV coefficient of about 1.5 would correspond to full pass-through of total DST-related costs. All estimates exceed this threshold.

B Additional Tables and Figures

Table B.1: Variable definitions and sources

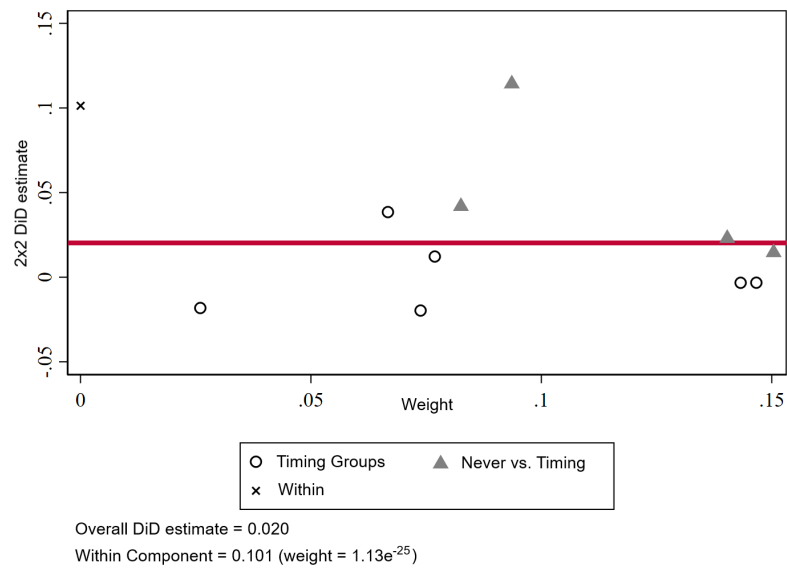
Variable	Definition	Source
FBA fee	Sum of storage fee, shipment fee and optional services fees. The shipment fee is the fee as given in the rate cards for a given weight, size and packaging type. The storage fee is calculated as the rate per volume multiplied by the maximum volume of the packaging type. Among the optional services, we only consider the bagging, labelling and taping fees.	FBA Rate Cards (Available at: sell.amazon.co.uk)
Inflation	Headline consumer price index, monthly	World Bank
Covid-19 index	Average of the four indices from the Oxford Covid-19 Government Response Tracker aggregated to monthly level. Set to zero for periods before 01/01/2020.	Oxford Covid-19 Government Response Tracker (OxCGRT)
Market share of the digital economy	‘Retail sale via mail order houses or via Internet’ divided by the total retail sales in the country (approximated by ‘Retail trade, except of motor vehicles and motorcycles’). Annual levels interpolated to monthly data using the monthly sales growth data.	Eurostat
FBA price	Lowest price for the new product offered by a seller using the Fulfillment by Amazon service. Data scraped at irregular time intervals, averaged to monthly.	Keepa.com
Amazon price	Price for the new product when Amazon is the seller, averaged to monthly.	Keepa.com
Average supplier count	The monthly average of the number of suppliers selling the (new, not used) product on Amazon.	Keepa.com

Table B.2: Fee analysis: Bacon decomposition

	β	Total Weight
Timing Groups	0.001	0.53
Never vs. Timing	0.041	0.47

Notes: Decomposition of the estimated treatment effect on FBA fees using the method by [Goodman-Bacon \(2021\)](#). Results correspond to the specification in Table 4, column (2). The table separates the overall estimate into contributions from comparisons between units treated at different times (“Timing Groups”) and comparisons between treated and never-treated units (“Never vs. Timing”). β denotes the average treatment effect within each group, and “Total Weight” indicates its contribution to the overall effect. *Data:* Hand-collected Amazon rate cards, Dec 2017–Feb 2022. See Table B.1 for details.

Figure B.1: Fee analysis: Bacon decomposition



Notes: Visual representation of the Bacon decomposition of the estimated effect of the DST on FBA fees. Results correspond to the specification in Table 4, column (2), and are based on the method by Goodman-Bacon (2021). *Data:* Hand-collected Amazon rate cards, Dec 2017–Feb 2022. See Table B.1 for details.

Table B.3: Amazon prices

	(1)	(2)
DST Rate	0.002 (0.001)	0.001 (0.004)
Controls	✓	✓
Cross-section FE	✓	
Time FE	✓	
Product-time FE		✓
R ²	0.964	0.981
Observations	16,346,688	13,359,305

Notes: Estimated effects of DST implementation on prices for products sold directly by Amazon. Column (1) uses a two-way fixed effects (TWFE) specification with product–country and time fixed effects. Column (2) compares prices of the same product across countries in the same month using product–time fixed effects. Both specifications include controls for inflation, the market share of the digital economy, COVID-19 policy stringency, and the standard VAT rate to account for the temporary VAT reduction in Germany. The sample covers 12 months before and after DST implementation. Standard errors are clustered two-way by country and by time. ***, **, * indicate significance at the 1%, 5%, and 10% levels. *Data:* See Table B.1.

Table B.4: Fee analysis: Robustness to clustering at country $\times$ product type

Estimator	TWFE		BJS	SA	DCDH	CS
Expl. Var.	DST Rate %			Dummy		
	(1)	(2)	(3)	(4)	(5)	(6)
DST	0.005 (0.004)	0.020 (0.013)	0.022*** (0.005)	0.050*** (0.002)	0.019*** (0.005)	0.033*** (0.012)
France						0.025** (0.011)
Italy						0.016 (0.015)
Spain						0.024** (0.009)
UK						0.068*** (0.022)
Controls	✓	✓				
Cross-section FE	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓
R^2	0.985	0.985		0.991		
Observations	11,730	11,730	10,760	11,730	11,730	11,730
Clustering	entry $\times$ prod. type	entry $\times$ prod. type	entry $\times$ prod. type	entry $\times$ prod. type	entry $\times$ prod. type	entry $\times$ prod. type

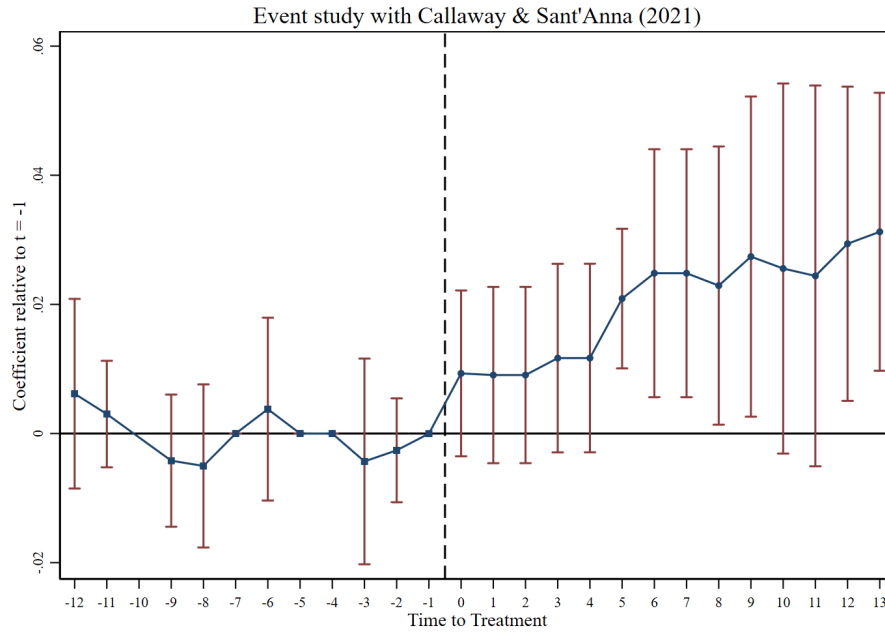
Notes: Replication of Table 4 with standard errors clustered at the country $\times$ product-type level (Media, Non-Media, Non-Media (CBS); 15 clusters in total). For the staggered DiD estimators in columns (5) and (6), inference uses a wild cluster bootstrap to account for the small number of clusters. ***, **, * indicate significance at the 1%, 5%, and 10% levels. *Data:* Hand-collected Amazon rate cards, Dec 2017–Feb 2022. See Table B.1 for details.

Table B.5: Number of marketplace sellers

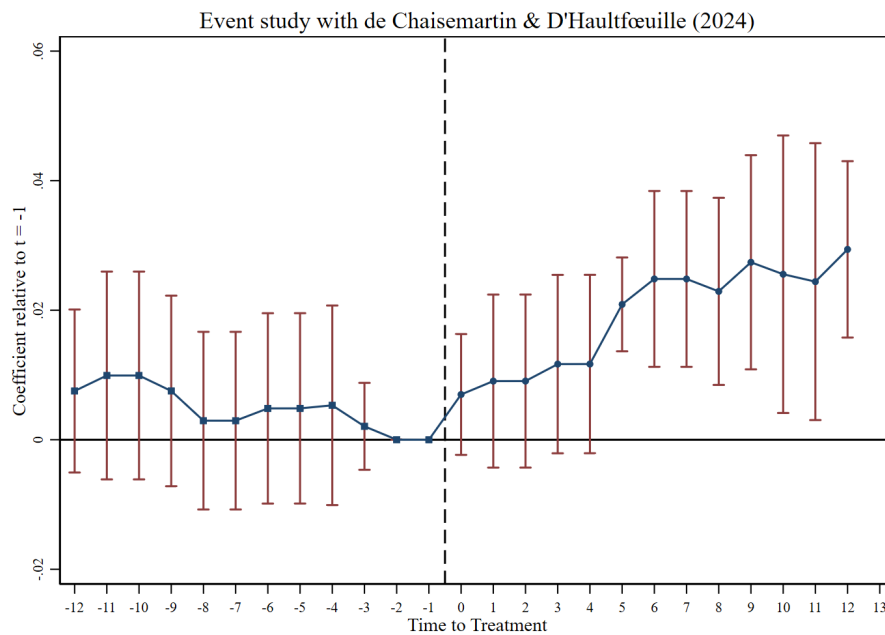
	France	Italy	Spain	United Kingdom
	(1)	(2)	(3)	(4)
DST $\times$ Post	-0.187 (0.622)	1.431*** (0.276)	-0.140 (1.057)	0.458 (0.280)
Controls	✓	✓	✓	✓
Product FE	✓	✓	✓	✓
Country FE	✓	✓	✓	✓
Time FE	✓	✓	✓	✓
R^2	0.716	0.728	0.717	0.708
Observations	11,049,186	11,943,635	11,111,835	13,904,048

Notes: Estimated effects of DST implementation on the number of sellers on the Amazon marketplace, using a DiD design. Each column compares a treated country (France, Italy, Spain, or the UK) to Germany as a control. All regressions include product, country, and time fixed effects and control for inflation, the market share of the digital economy, and a COVID-19 policy stringency index. Standard errors are clustered two-way by product and by month. ***, **, * indicate significance at the 1%, 5%, and 10% levels. *Data:* See Table B.1.

Figure B.2: Fee analysis: Other event study estimators



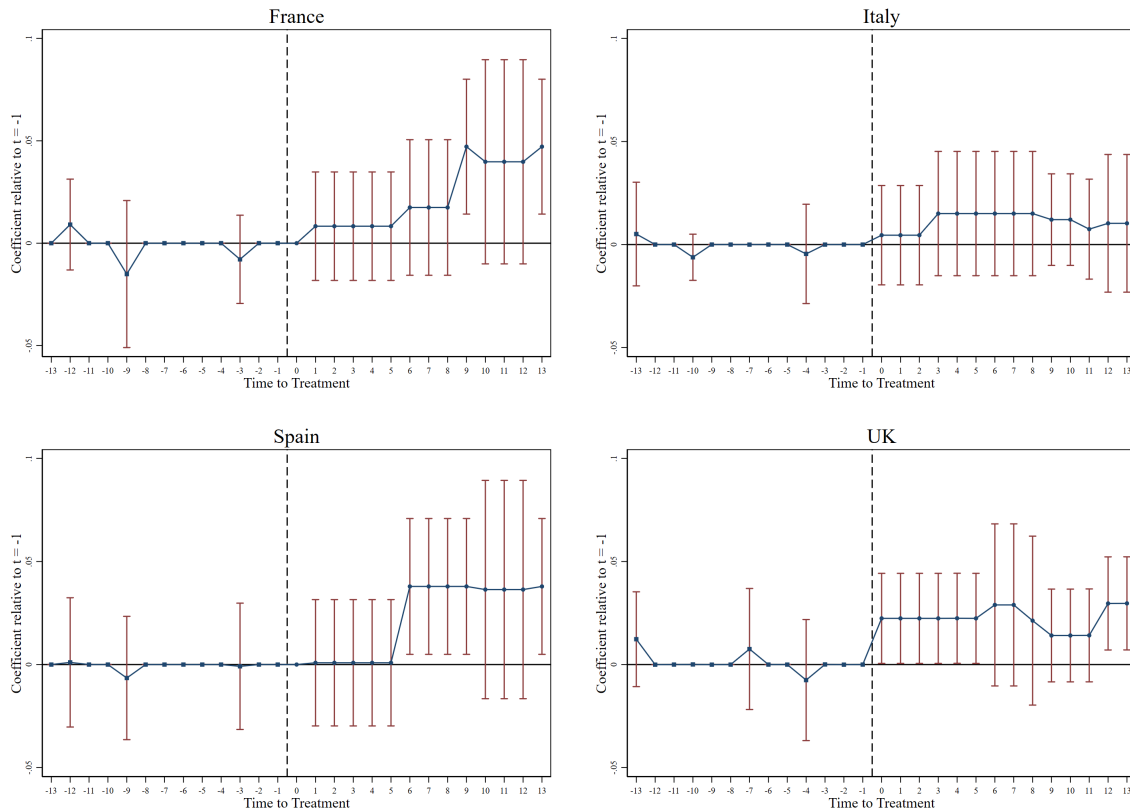
(a) Callaway–Sant’Anna (2021)



(b) de Chaisemartin–D’Haultfoeulle (2024)

Notes: Event study estimates of the effect of DST implementation on FBA fees across alternative estimators. Panel (a) reports results based on the estimator by Callaway and Sant’Anna (2021) using the short-gap specification. Panel (b) reports results based on the estimator by De Chaisemartin and d’Haultfoeulle (2024). Both panels show 95% confidence intervals. Standard errors clustered at the cross-sectional identifier in both panels. Only standard-sized products are included. Observations are weighted by frequency in the price data. Monthly values are imputed by assuming constant fees between published rate cards. *Data:* Hand-collected Amazon rate cards, Dec 2017–Feb 2022. See Table B.1 for details.

Figure B.3: Fee analysis: Event studies by country



Notes: Event study estimates of the effect of DST implementation on FBA fees, separately for each country. Estimates are based on the estimator by [Callaway and Sant'Anna \(2021\)](#), based on observations in the 13 months before/after DST introduction. 95% confidence intervals. Standard errors clustered by cross-sectional identifier. Only standard-sized products are included. Observations are weighted by frequency in the price data. Monthly values are imputed by assuming constant fees between published rate cards. *Data:* Hand-collected Amazon rate cards, Dec 2017–Feb 2022. See Table B.1 for details.

Table B.6: Price analysis: DiD results without interpolation

	(1) France	(2) Italy	(3) Spain	(4) United Kingdom
$FBA_{is} \times DST_t$	0.023* (0.013)	0.005 (0.008)	0.021** (0.008)	0.020** (0.008)
Product-by-seller-type FE	✓	✓	✓	✓
Time FE	✓	✓	✓	✓
R ²	0.96	0.96	0.96	0.96
Observations	1,606,985	2,062,854	1,700,448	2,690,171

Notes: Estimated effects of DST implementation on consumer prices, using a DiD specification. The outcome is the logarithm of the price for products sold by third-party sellers using FBA, compared to prices of identical products sold by Amazon directly. All columns include product-by-seller-type and time fixed effects. Only standard-sized products are included. Data for nine months before and after DST introduction without interpolation. Standard errors are clustered two-way by product category and by month. ***, **, * indicate significance at the 1%, 5%, and 10% levels. *Data:* Keepa.com. See Table B.1 for details.